Media Wall & Dining Wall: Build Guide

Version 42 · for Emma and Dave · sizes in cm



What you're making

Ch.	What	Time	£
1	The media wall base	2–3 days	517–1,012
2	The oak worktop	½ day	250–450
3	The oak uprights and shelves	2–3 days	420–765
4	The side-wall run and the PC shelf	1 day	60–130
5	The dining cabinet and table	3–4 days	906–1,535
6	The bar	1 day	120–200
7	The glass case	½ day	90–170
8	The record table	½ day	170–210
9	Bought and fitted: shelves, cabinets, the door shelf	1 day	195–300
A	The network	2 h	51–70
B	Cables and the socket	2–3 days	165–345
C	Paint	2–3 days	220–540
D	Lighting	1 day	288–795
E	Connect and finish	1 day	433–725
	Also on the list (the basket, the rug, the ottoman…)		489–998
	Everything	about 17–22 days	**4,374–8,245**

Each chapter: **Buy** (what that thing needs), **Cut** (every part drawn to scale with its size) and **Build** (numbered steps, then a check). The merchant's full cutting list and the sheet layouts are at the back.

Start here

Five rules

- **Check every wall before you drill** with the detector; never straight above, below or beside a socket or switch.
- **Measure twice, cut once.** Walls are rarely square: cut to the smallest measurement.
- **Two people for anything over 15 kg:** the uprights, the worktop, the table, the TV.
- **Electrics:** isolate, lock off and prove dead every time. Anything new is an electrician's (chapter B's stop list).
- **Dry-fit everything** before glue or screws.

The order of work

1. **A** · The network
2. **B** · Cables and the socket
3. **C** · Paint
4. **Start here** · Mark the heights
5. **1** · The media wall base
6. **2** · The oak worktop
7. **3** · The oak uprights and shelves
8. **4** · The side-wall run and the PC shelf
9. **5** · The dining cabinet and table
10. **D** · Lighting
11. **E** · Connect and finish
12. **6–9** · The bar, the glass case, the record table, the fitted pieces

Measure and mark

1. The alcove's width at the floor, 76, 134, 155 and 196 cm and at the ceiling: cut everything to the smallest.
2. The floor's level across the alcove, and the chimney breast's flatness with a long level.
3. The door's swing, and the sofa: overall width, depth with the return, back height.
4. Count the LPs and games (PS2, Xbox 360, Xbox One), and measure ten cases of each.
5. Make a cardboard template of the alcove at worktop height: walls are rarely square.

Set up a datum

6. Find the floor's highest point across the alcove (you measured its level in Start here). Every height in this plan is measured from there.
7. Stand the laser in the middle of the room, level it, and mark a line at exactly 100 cm above that high point on the alcove's back wall, both side walls and the chimney breast.
8. Measure every other height up or down from the 100 line, not up from the floor: the floor is the thing that isn't level.

The alcove's heights

9. Put a strip of masking tape down each side wall and the back wall of the alcove and mark these heights on it:
10. 73: the tops of the frames and bay 3 (the worktop's underside).
11. 76: the worktop's top.
12. 130 and 134: the Series X shelf, underside and top.
13. 153 and 155.4: the record ledge and the wrap ledge, underside and top.
14. 192 and 196: the high shelf, underside and top.
15. Carry the 153 and 155.4 lines along the side wall to 250 cm out from the alcove's back wall, for the wrap ledge and the PC shelf.

Across the back wall

16. Mark the uprights' positions from the side wall: the left one 50.5–55, the right one 181.5–186. Turn the laser to its vertical line and draw both lines from the worktop line to the ceiling.
17. Mark the base from the side wall along the floor: filler 0–3, then bay 3 3–63, bay 2 63–123, bay 1 123–183, filler to 186.

On the chimney breast

18. Find the breast's centre: 58.75 cm from its alcove-side corner. Draw it plumb from the floor to 170.
19. Mark the “now playing” ledge's top at 104, level, and its two concealed brackets 6 cm in from each side of the breast.
20. Mark the glass display case's bottom edge at 137 and its top at 222, 90 wide, centred on the line; its oak cleat's top edge at 212. (The dining table isn't here: it's in its cabinet by the stairs, chapter 5.)

Check for cables

21. Run the detector along every line where you'll drill: the cleats, the spacer blocks, the upright brackets, the ledge's brackets, the PC shelf and, on the stairs wall, the cabinet's two wall brackets.
22. Mark anything it finds in red, and compare with your own photos from chapter B. Move a fixing 5 cm rather than drill near a cable.

Check before moving on: Every figure within 1 cm of the plan. If not, send the numbers and the drawings are redone first.

Check before moving on: Every line level end to end, and the uprights' lines plumb.

Measuring sheet: 11 things to check before you buy

One session with a tape: measure each, write it in, send us the page.

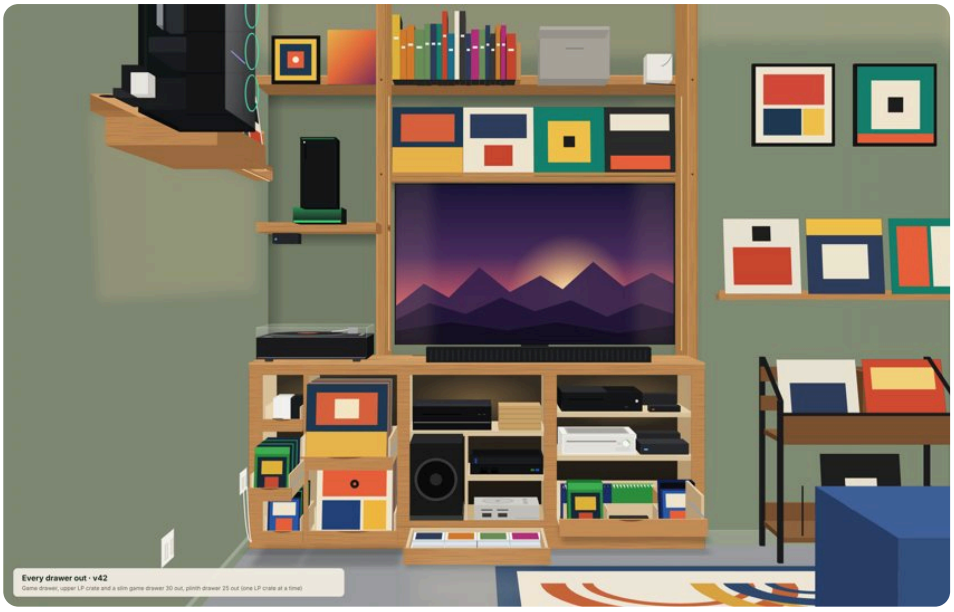
What	We drew	Why it matters	Yours (cm)
The TV's remote receiver, bottom centre of the screen's frame The TV between the uprights	soundbar top 82, in front of the stand's foot	0.6 over the frame's edge: sit the bar on the worktop 2 cm proud of the stand's foot, not against it, so the remote's beam from the sofa clears it; if a button stutters, pair LG's Magic Remote (it talks to the TV by radio)	
The TV's HDMI plugs, with the screen 2.15 from the corner post The TV between the uprights	behind the screen	plugs stay behind the panel; if a stiff lead pushes out, use 90° HDMI adapters	
Xbox 360, original, + RetroTINK-6X CE Bay 1, Xbox (BESTÅ frame on the 9 cm plinth)	16 tall row	7.7 above; 25.9 beside it for the RetroTINK: measure it when it arrives	
Vibration: the consoles and the deck share the base with the sub Bay 2, PlayStation and the sub (BESTÅ frame on the plinth)	the sub on its riser, clear of the divider and the frame; the deck on isolation feet	the 360 is a bay away, the PS2 and PS1 a divider away: play a disc game and a record at film volume; if anything skips, turn the sub down for that source, or put it back under the dining wall	

The side-wall socket beside the slim drawers Bay 3: the LP crates and the slim game column	the middle slim drawer pulled out passes 5.6 from the wall	the plugs must stand less than that off the wall: flat (angled) plugs, leads down; measure yours	
The two doors and their hinges The compact dining cabinet (the plan)	a full-height stainless piano hinge each, screwed every 10 cm into a solid oak edge	the hinge's load isn't published: hang the table door, open it flat, and check its far end doesn't drop before you load the cabinet	
Stair side: the stair's foot and the door The compact dining cabinet (the plan)	the stair box's face at -41; the door's edge reaches -44	seated, the backs reach 8.7 into the door's swing: keep the door shut while you eat. Measure a chair's depth: 52.3 or less and the door clears	
Its stairs end Your black shelf on the stairs wall (a fixed 10 cm gap)	an earlier look at your photo put it about 41 past the stair box	35 past it: if you measure 41 there, the shelf is about 194 long; the gap stays 10	
The shelf over the door Your picks (in the plan)	the door's 198 and a 7 architrave	11.3 over two 15-high boxes; check your architrave	
The step drawers Your picks (in the plan)	a riser 19.5 under a 2.2 tread, the stairs 59 wide	each tread needs a bearer under it once its riser comes out: a joiner's job, or build them into the new stairs	
Its height The storage ottoman (replaces your blue cord one)	the sofa's seat about 45	a footrest level with the seat; ISTOOLS don't publish it: check	

1 · The media wall base

The plinth, two BESTÅ frames and bay 3's carcass, with the drawers, the LP crates and the oak face frame.

2–3 days · You · about £517–1,012



1 · Buy

Qty	What	Where	£
2	**IKEA BESTÅ frame, 60 × 40 × 64** Two IKEA BESTÅ 60×40×64 frames, white stained oak effect, for bays 1–2	IKEA	70–200
1	**Plinth frame, 18 mm ply** Plinth for bays 1–2: 18 mm ply ladder frame 120 × 38 × 9 cm with a clear bay under bay 2, plus shims	Timber merchant	15–30
1	**Plinth drawer: oak front, ply box, runners** Hidden plinth drawer: oak front 54 × 6 with a finger-pull groove underneath, 12 mm ply box 52 × 34 × 5, 300 mm full-extension runners	Timber merchant	30–50
1	**Bay 3 carcass, 18 mm birch ply** Bay 3 carcass: 18 mm birch ply, 60 × 40 × 73 cm from the floor, connector bolts, and a 12 mm birch-ply divider 38 × 69.4 that splits it into the slim game column and the LP column	Timber merchant	50–90
2	**Face-out LP crate: ply, oak lip, runners** Two face-out LP crates, 34.9 × 40 (a 33 cm trough): 12 and 9 mm birch ply cut to size, 15 mm oak lips, raked rests, felt, two pairs of 400 mm runners rated 68–70 kg	Timber merchant	120–200
3	**Slim game drawer: ply, oak lip, runners** Three slim drawers for the game column, 15.3 × 40: 6 mm birch ply sides and backs, 12 mm bases, oak lips, felt, three pairs of 400 mm runners (light duty is enough); a 2 m USB-C lead and a grommet for the pads drawer	Timber merchant	70–130
4	**IKEA BESTÅ shelf, 56 × 36** BESTÅ shelves, 56 × 36 cm (20 kg max each), two each for bays 1 and 2	DIY shop or online	12–32
1	**Sub riser and ply divider** The sub in bay 2: an isolation riser for it (dense foam or sorbothane, about 22 × 28 × 3), 12 mm birch ply for the divider (35.3 × 38) and the part shelf (32.2 × 36), four shelf pins, felt pads	Timber merchant	25–45

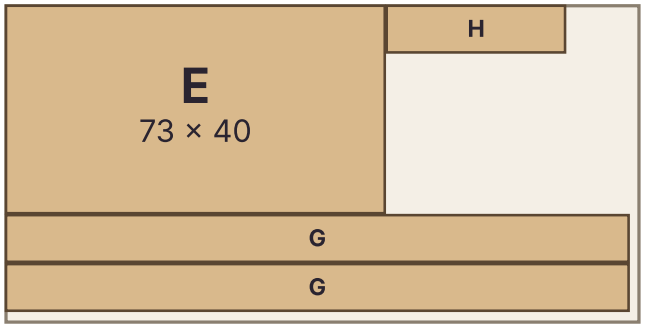
1 set	**Anti-tip brackets and spacers** Anti-tip brackets and 4 cm spacer blocks: both frames and bay 3	DIY shop or online	15–25
1	**Game drawer: ply, soft-close runners** Game drawer kit: 12 mm birch ply cut to size, 350 mm full-extension soft-close runners (35–49 kg), oak lip, follower block, felt	Timber merchant	50–90
1	**Oak face frame, 186 cm** Oak face frame, 20 mm, 186 cm for the base	Timber merchant	60–120
	Total		**517–1,012**

2 · Cut

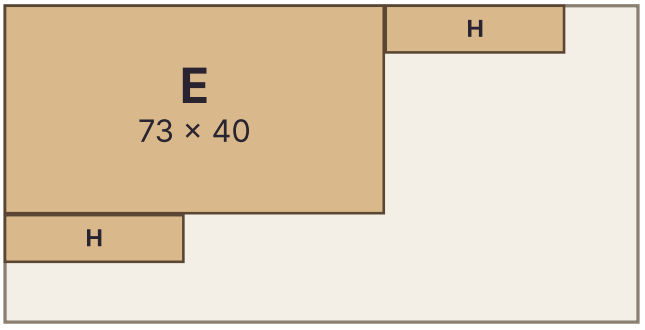
Solid oak, hardwax oil: each piece to scale



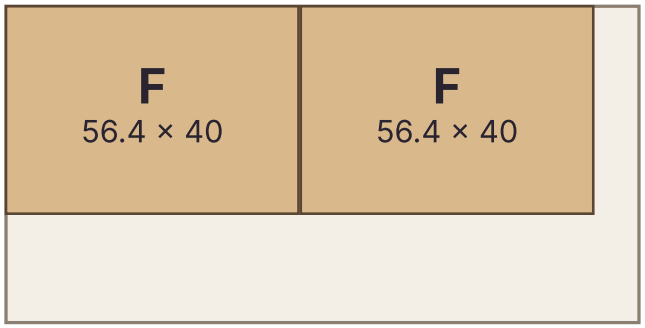
Birch ply, 18 mm: board 1 of 3, 122 × 61



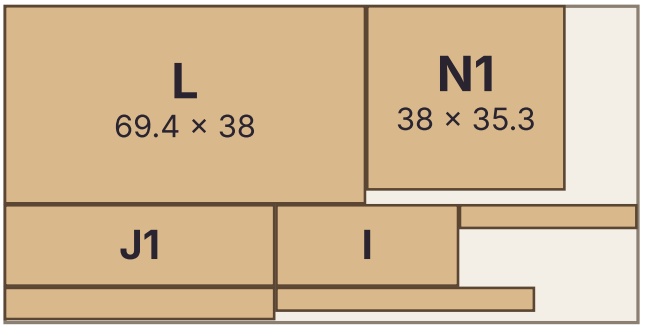
Birch ply, 18 mm: board 2 of 3, 122 × 61



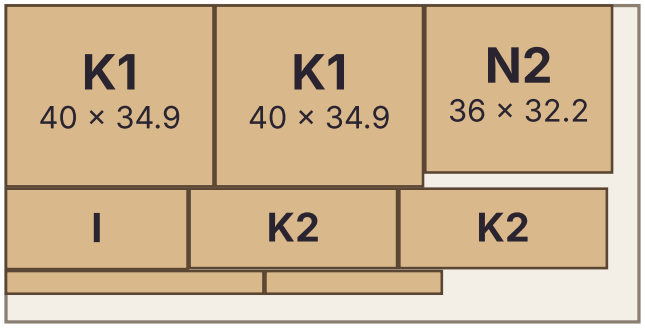
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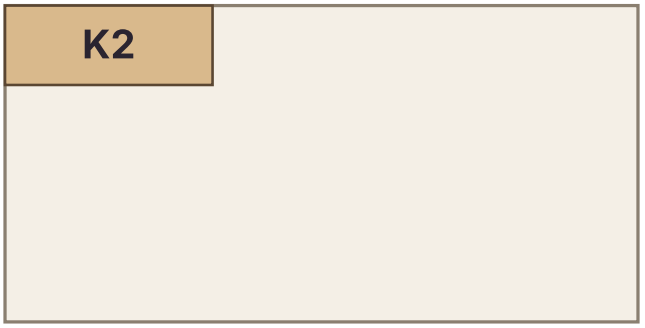
Birch ply, 12 mm: board 1 of 3, 122 × 61



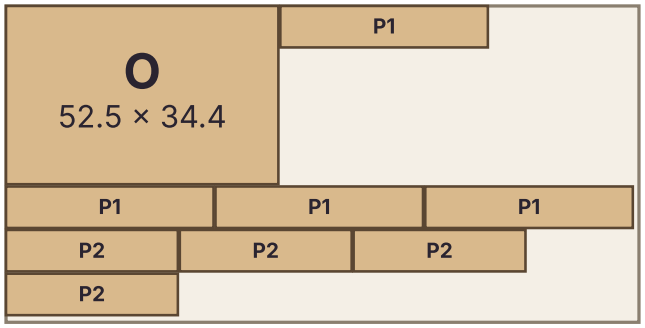
Birch ply, 12 mm: board 2 of 3, 122 × 61



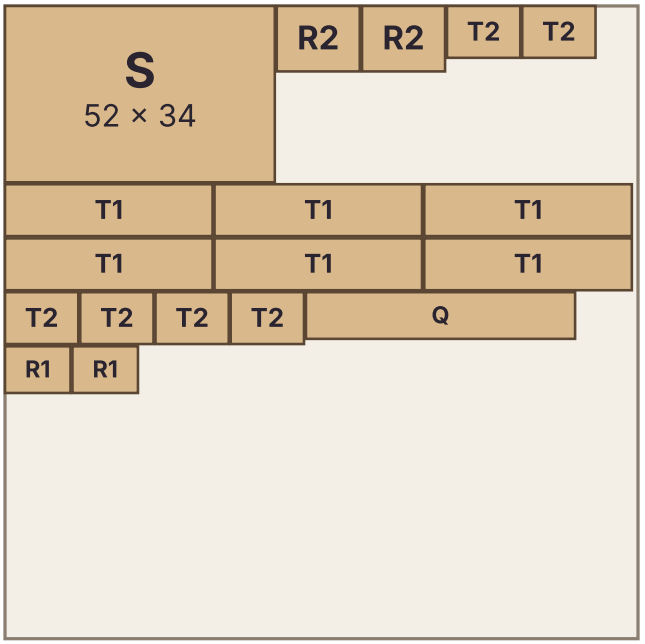
Birch ply, 12 mm: board 3 of 3, 122 × 61



Birch ply, 9 mm: board 1 of 1, 122 × 61



Birch ply, 6 mm: board 1 of 1, 122 × 122



Part	What	Size (cm)	Qty	From	Notes
A	Face frame	186 × 73 overall, 2 thick, stiles about 5 wide	1	Solid oak	End stiles scribed to the walls
B	Drawer fronts	game drawer 56.8 × 6.5 × 2 · plinth drawer 54 × 6 × 2	1 · 1	Solid oak	Finger pull cut underneath
C	LP crate lips · slim drawer lips	36.3 × 5 × 1.5 · 16.5 × 5 × 1.5	2 · 3	Solid oak	Cut to the face frame's openings, less 3 mm
D	Follower blocks · Xbox One lift block	19 × 12 × 2 · 15 × 12 × 2	2 · 1	Solid oak	Offcuts, felted underneath
E	Bay 3 sides	73 × 40	2	Birch ply	The record carcass, standing on the floor
F	Bay 3 top and bottom	56.4 × 40	2	Birch ply	Between the sides; back left open to the channel
G	Plinth rails	120 × 9	2	Birch ply	Front and back of the ladder frame under bays 1–2
H	Plinth cross-pieces	34.4 × 9	3	Birch ply	At both ends and under the bay 1–bay 2 joint, so the bay under bay 2 stays clear for the drawer
I	Game drawer sides	35 × 15.5	2	Birch ply	
J	Game drawer back · front panel	51.9 × 15.5 · 51.9 × 6	1 · 1	Birch ply	
K	LP crate bases · slim drawer bases	34.9 × 40 · 15.3 × 40	2 · 3	Birch ply	The sides and ends stand on them
L	Bay 3 divider	69.4 × 38	1	Birch ply	17.8 in from the left side, the full height inside
M	Plinth drawer sides · ends	34 × 4.4 · 49.6 × 4.4	2 · 2	Birch ply	A 52 × 34 × 5 box

N	Bay 2 divider · part shelf	35.3 × 38 · 32.2 × 36	1 · 1	Birch ply	The sub's column on the left; the PS2 over the PS1 on the right
O	Game drawer base	52.5 × 34.4	1	Birch ply	In a 6 mm groove
P	LP crate sides · ends	40 × 8 · 33.1 × 8	4 · 4	Birch ply	Two of each per crate: a 33 cm trough
Q	Game drawer cross divider	51.9 × 9	1	Birch ply	13 cm back from the front
R	Game drawer bin walls · bin floors	12.6 × 9 · 16 × 12.6	2 · 2	Birch ply	Floors raked so the cases lean about 13°
S	Plinth drawer base	52 × 34	1	Birch ply	Screwed under the box
T	Slim drawer sides · backs and fronts	40 × 10 · 14.1 × 10	6 · 6	Birch ply	Three boxes 15.3 wide, 14.1 inside

Sizes are finished sizes, length × width × thickness; a 4 mm saw cut between pieces. Ply parts can be cut by the merchant from these layouts.

3 · Build

Tools

- Drill-driver with pilot and countersink bits
- SDS drill with a 7 mm masonry bit
- Spirit level, 60 and 120 cm
- Combination square
- Clamps
- Tape measure

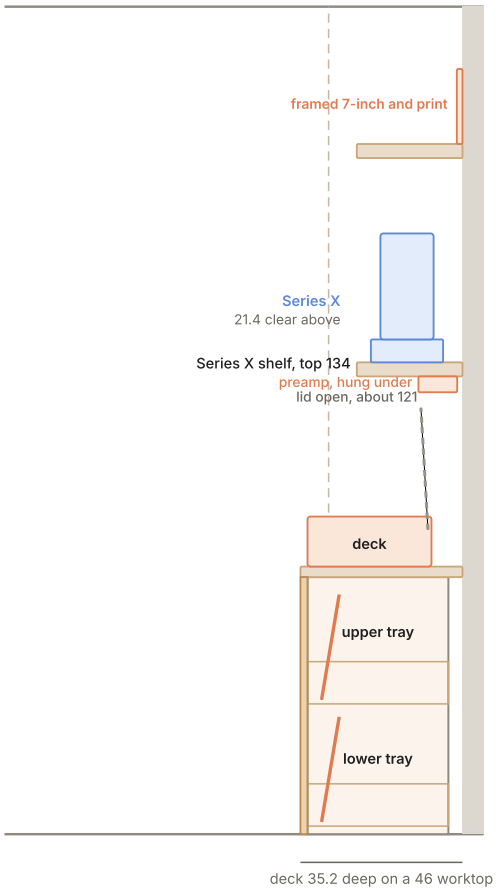
Fixings

- 4 × 40 and 4 × 50 mm wood screws
- 5 × 60 mm screws and brown wall plugs
- Wood glue

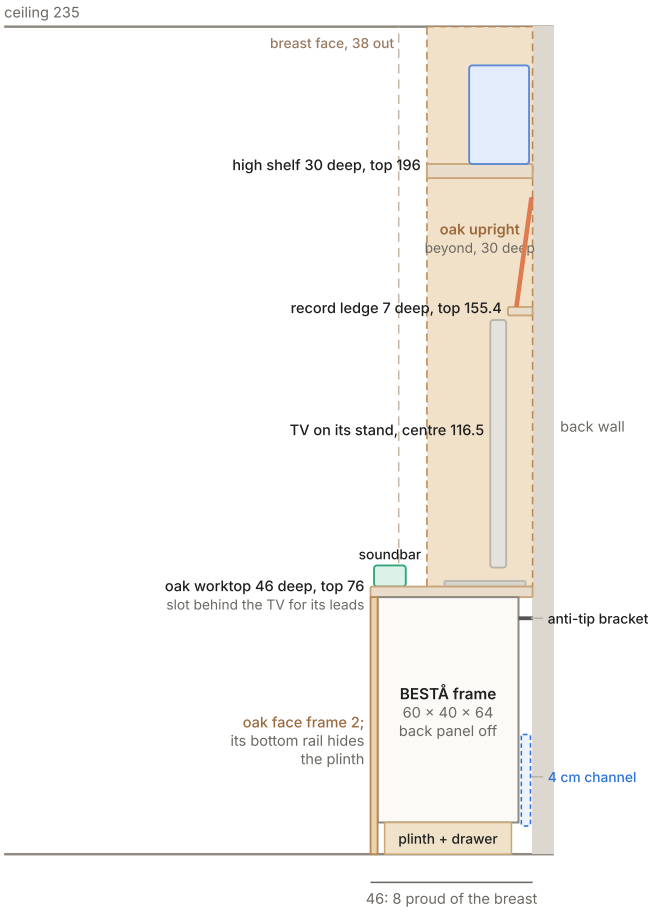
How the wall goes together, layer by layer

Side sections to scale, room on the left. Dashed: the upright and the breast's face beyond the cut, and the deck's lid open

B · Through the turntable zone



A · Through the TV



The plinth under bays 1 and 2

1. Lay the two 120 × 9 rails on edge, 38 cm apart outside, and stand the three 34.4 × 9 cross-pieces between them: one at each end and one in the middle, at 60.
2. Pre-drill, glue and screw through the rails into each cross-piece end with two 4 × 40 screws.
3. Measure both diagonals: when they're equal the frame is square. The plinth is 120 × 38 × 9. The opening between 0 and 60 (the end nearer the side wall) stays clear: it's the hidden drawer's bay, under bay 2.
4. Stand it in the alcove 63 cm from the side wall (bay 3 stands on the floor between it and the wall) and 4 cm off the back wall. Level it with shims, side to side and front to back.

The BESTÅ frames

5. Assemble both frames to IKEA's instructions, but leave the back panels off: the open backs are how the cables and the sub's port breathe.
6. Join the two frames side by side with IKEA's connectors.
7. Stand them on the plinth, their backs flush with its back edge, so their fronts sit 2 cm proud of the plinth (the face frame covers the step).

Bay 3, the carcass for the LP crates and the slim column

8. While the sides are flat, mark the runner lines on their inner faces: on the right side 2.3 and 37.0 cm up from the bottom edge, for the two LP crates; on the left side 2.3, 25.1 and 47.9, for the three slim drawers.

9. Stand the bottom panel between the two sides, flush with their bottom edges, and the top panel flush with their top edges. Pre-drill, countersink, glue and screw with four 4 × 50 screws a joint.
10. Check the diagonals for square. The carcass is 60 × 40 × 73, open at the back.
11. Stand it in its place, 3–63 from the side wall, its top level with the frames' tops at 73. Shim under it if the floor drops away.
12. Clamp it to bay 2 and fit two connector bolts at the front and two at the back.

Fix it to the wall

13. Screw a 4 cm spacer block to the back wall behind the top of each bay, at 68 high, with plugs.
14. Push the whole base back against the blocks.
15. Fit two anti-tip brackets per bay through the blocks into the wall. Do this before anything heavy goes in: a full LP crate pulled out is about 30 kg in front of the unit.
16. Lay a level across all three tops: within 1 mm over 186 cm. Shim and re-check.

The cable channel

17. Lay the flat two-compartment trunking on the floor in the 4 cm channel behind the base: mains on the wall side, data on the other.
18. Run the leads in now: the two mains leads from the side-wall socket enter through the face frame's notch and run back in the 3 cm filler beside bay 3; the Cat6 and Deco B's lead come down the right-hand corner.
19. Keep the channel behind bay 2 empty: the sub's port breathes into it. Switch 2 and the retro 6-way stand on edge behind bay 1.

Tools

- Drill-driver
- Router with a 6 mm straight bit (the game drawer's base groove)
- Combination square
- Spirit level
- Clamps
- Mitre saw or tenon saw

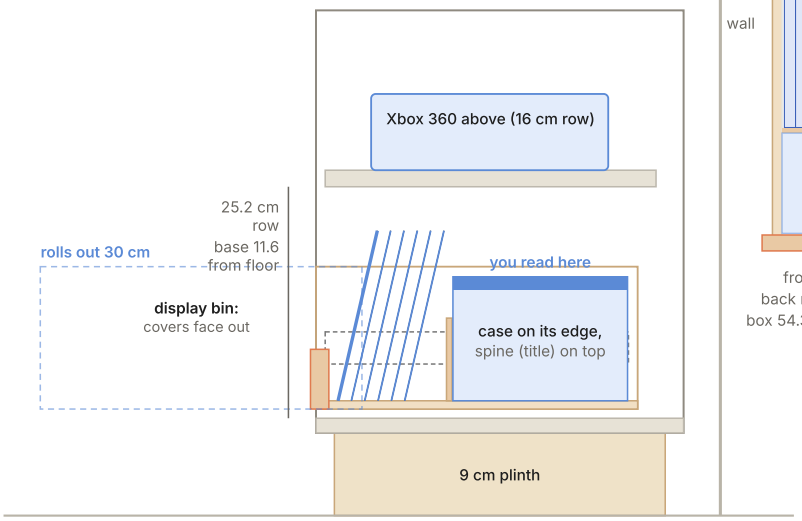
Fixings

- Runners: two pairs of 400 mm (68–70 kg) for the LP crates, three pairs of 400 mm light-duty for the slim drawers, one pair of 350 mm soft-close, one pair of 300 mm
- 3.5 × 16 mm screws for the runners (every hole)
- Four shelf pins for the part shelf
- Two small steel angle brackets for the divider
- Panel pins, wood glue, felt

The game drawer: covers on show at the front, every title readable from above at the back

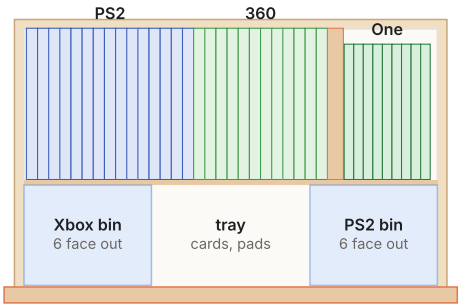
Bay 1, bottom row. Section left, plan right. Nothing stands behind anything else

Side section through bay 1



box 35 cm deep on 350 mm full-extension soft-close runners
5 cm free behind the box, then the 4 cm cable channel
bins: 19 cm cases leaning 13°, about 18.5 cm tall

Top view, drawer pulled out

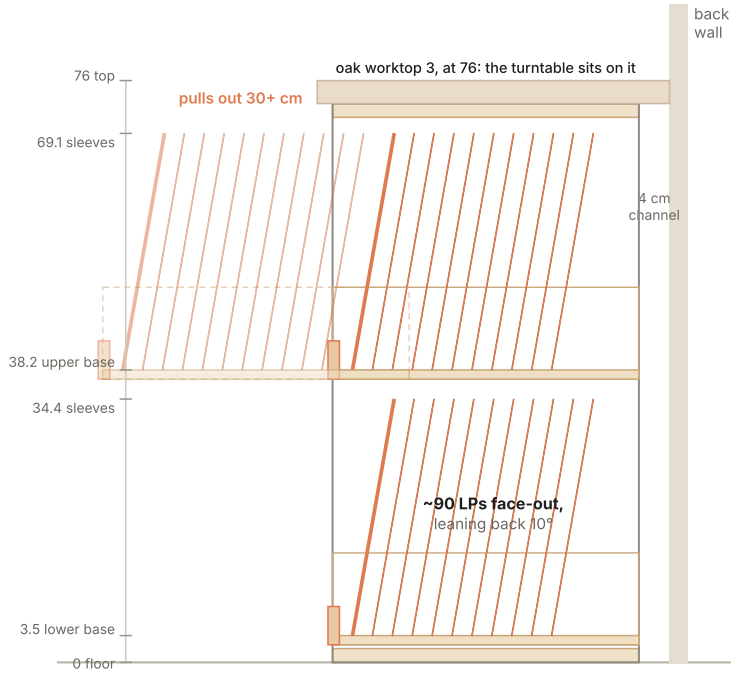


front: oak lip with finger pull, matching the record drawer
back row: 15 PS2 + 12 Xbox 360 + follower block + 9 Xbox One
box 54.3 × 35 cm outside · 51.9 cm inside · 16 cm bins, 18.7 cm tray

Bay 3: two LP crates under the turntable, a slim game column beside them

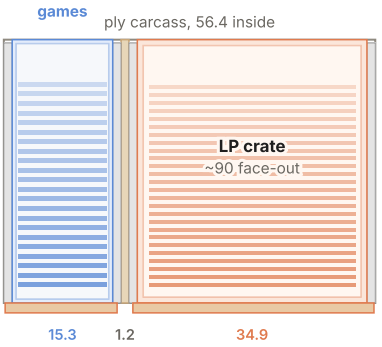
Section left through the LP crates, plan right. Pull a crate and flip through the covers like a record shop browser; the slim drawers flip the same way for games

Side section through bay 3



carcass 18 mm birch ply, 60 × 40 × 73, back open to the channel · crates 40 deep on 400 mm runners
12 mm base, 9 mm sides, 5 cm oak lip · 2.6 and 2.1 cm finger room (top panel at 71.2)

Bay 3 from above



LP crate 34.9 × 40, a 33.1 trough: LPs snug, no dead space
slim drawers 15.3 × 40, 14.1 inside: DVD-size cases face-out
12 mm ply divider · 12.7 mm runners each side
an LP crate is about 30 kg full: open one at a time

The console shelves

20. Bay 1, from the top: rows of 16 (Xbox One), 16 (Xbox 360), then the game drawer's 25.2. Put the shelves on the pin holes nearest those heights.
21. Bay 2: the PS3's row of 23.7 at the top, on a full-width shelf. Below it, the PS2 and the PS1 sit beside the sub (next part).

Bay 2's sub column

22. Stand the 12 mm divider on the bay's floor, 23.4 cm from the bay's left inner face, tight under the PS3 shelf. Check it's plumb.
23. Fix it with one small angle bracket at the bottom (into the frame's bottom panel) and one at the top (into the PS3 shelf's underside).
24. Drill two 5 mm pin holes in the divider's right face at the height of the frame's pin holes for the PS2/PS1 shelf. Fit the part shelf on four pins.
25. Pair the sub with the soundbar now: its PAIR button is on the back, and it's hard to reach once it's in.
26. Lay the isolation riser on the column's floor, its front edge flush with the frame's front.
27. Stand the sub on it with the grille facing the room, centred: 1 cm to the divider, 1 cm to the frame's side. Slide it forward until its front is flush with the face frame's front (2 cm proud of the frame). Run its lead out of the back.
28. Check it touches nothing but the riser. If it does, a sheet of felt between, or nudge it.

The game drawer (bay 1, bottom row)

29. Fit the 350 mm runners to the frame's sides, level, so the drawer's base will sit about 1 cm above the frame's bottom panel. Drill pilot holes: BESTÅ sides are particleboard. If a screw won't hold, glue a 12 mm ply packer behind the runner.
30. Rout a 6 mm deep groove 1 cm up from the bottom on the inner faces of the sides, back and front panel, 9 mm wide, for the base.
31. Glue and screw the box together round the base: outside 54.3 × 35. Check the width against the runner maker's figure (usually the opening less 25.4 mm) before the glue sets.
32. Fit the cross divider 13 cm back from the front, and the two bin walls 16 cm in from each side. Glue in the bin floors with a slight back-rake, so the cases lean about 13°.
33. Felt the base and make the follower blocks.
34. Fit the runners' inner members to the box, hang it, and check it runs true empty.

Bay 3: the divider, the LP crates and the slim drawers

35. Stand the 12 mm divider in the bay, its left face 17.8 cm from the left side's inner face, the full height inside. Check it's plumb and parallel to the side at front and back (measure both), then fix it with two small angle brackets top and bottom.
36. Mark the runner lines on the divider's faces: on its right face 2.3 and 37.0 (the crates), on its left face 2.3, 25.1 and 47.9 (the slim drawers), matching the lines on the sides.
37. Screw the crates' 400 mm runners to the right side and the divider's right face; the slim drawers' to the left side and the divider's left face. Use every hole: a full crate is about 30 kg.
38. For each LP crate, glue and pin the two 40 × 8 sides and two 33.1 × 8 ends on top of the 34.9 × 40 base. The trough is 33 inside: snug for 31.4 sleeves, like a record shop's crate.
39. Glue a raked rest (an oak offcut) at the back of each crate, so the sleeves lean back about 10°. Felt the trough.
40. For each slim drawer, glue and pin the 6 mm sides, back and front onto the 15.3 × 40 base: 14.1 inside, for 13.5 cm game cases face-out. In the top one, glue a 6 mm strip down the middle (two pads on edge each side) and drill a 20 mm hole in the back for the USB-C lead's grommet.
41. Hang each box and run it in and out empty, then with ten records or ten games. Feed the charging lead from a USB plug on the 10-gang down the channel, with a loop of slack so the drawer opens fully.

The hidden plinth drawer

42. Make the box: the two 34 × 4.4 sides, the two 49.6 × 4.4 ends between them, the 6 mm base screwed underneath. It's 52 × 34 × 5.
43. Pack the plinth's inner faces under bay 2 with ply strips so the 300 mm runners sit the box's width plus their allowance apart, and fit the runners with the box 1.6 cm off the floor.
44. Felt the runners: the drawer sits under the sub.

The face frame and the drawer fronts

45. Cut the stiles (about 5 wide, and one 2 wide over bay 3's divider) and the rails from 2 cm oak. The bottom rail covers the plinth, 9 tall under bays 1–2, with an opening 54 × 6 for the plinth drawer, 1.6 up, under bay 2.
46. Scribe the two end stiles to the walls, then glue and pin the frame to the frames' fronts and bay 3. Notch the bottom rail's left end for the two socket leads.
47. Screw the oak lips onto the game drawer, the two LP crates, the three slim drawers and the plinth drawer from inside each box, with a 2–3 mm gap all round to the face frame.
48. Cut each finger pull underneath its lip: a cove routed along the back of the bottom edge.

Check before moving on: The tops level within 1 mm across 186 cm, and nothing rocks.

Check before moving on: Every drawer runs smoothly, the two crates clear each other by 2 cm, and the slim drawers by 2.8.

2 · The oak worktop

One solid oak worktop scribed to the alcove, over the base.

½ day · You + a helper · about £250–450

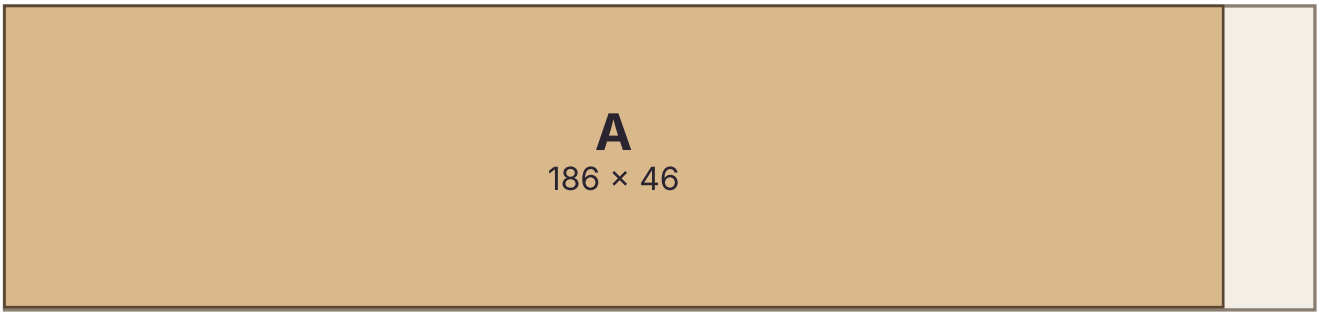


1 · Buy

Qty	What	Where	£
1	**Solid oak worktop, 300 × 62** One solid oak worktop, 300 × 62 cm, full stave: the alcove top (186 × 46), chamfered front; keep the offcut	Timber merchant	250–450
	Total		**250–450**

2 · Cut

Solid oak, hardwax oil: each piece to scale



Part	What	Size (cm)	Qty	From	Notes
A	Worktop	186 × 46 × 3	1	Solid oak	From one 300 × 62 full-stave worktop; chamfered front

Sizes are finished sizes, length × width × thickness; a 4 mm saw cut between pieces. Ply parts can be cut by the merchant from these layouts.

3 · Build

Tools

- Hardboard or cardboard, 190 × 50, for the template
- Compass or a pencil on a block, for scribing
- Track saw or circular saw with a guide
- Jigsaw
- Router with a straight bit
- Random-orbit sander, 120 and 180 grit
- Drill-driver
- A helper

Fixings

- 4 × 30 mm screws with washers (9)
- Hardwax oil

Make a template

1. Cut the hardboard to about 190 × 50 and push it into the alcove at worktop height, its back edge against the back wall.
2. Set the compass to the widest gap at one end and run it down the side wall, so the pencil copies the wall's line onto the board. Cut to the line. Do the other end the same way.
3. Mark the front edge at 46 cm from the back wall all along. The template is now the worktop's exact shape.

Cut the oak

4. Lay the board best face up, grain running along its length. Lay the template on it and draw round it.
5. Cut just outside the line with the track saw, then sand or plane to the line.
6. Keep the offcut flat and oiled with the rest: it's spare oak for any repair.
7. Chamfer the worktop's top front edge with a 3 mm chamfer, sanded by hand.

Cut the openings

8. The TV slot: 3 cm wide, from 1 to 4 cm in from the back edge, running 78.25–158.25 from the side wall. Clamp a batten as a guide and rout it in passes, or drill each end and jigsaw between. Push the brush strip in.
9. The grommet hole behind the deck: 35 mm, its centre 2.5 cm from the back edge and 25.25 from the side wall.

Finish and fix

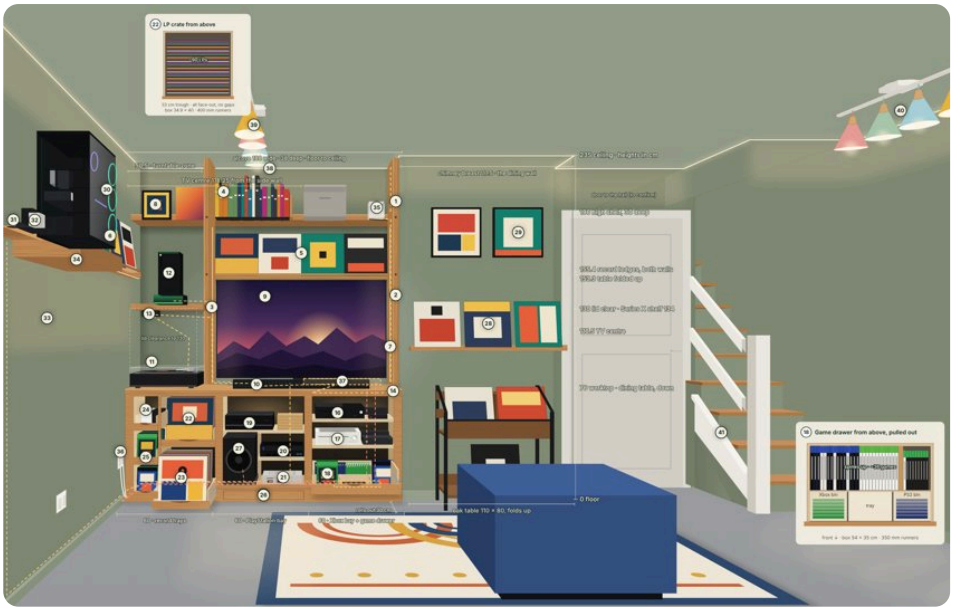
10. Sand to 120, then 180. Oil every face: top, underside, both ends and the cut edges of the slot. Two coats, a night between them.
11. Two of you lift it in and push it back to the wall. It should touch both side walls along their length.
12. Inside each bay, drill three 5 mm holes up through the top panel, elongated front to back to about 12 mm long, so the oak can move with the seasons.
13. Screw up through them into the worktop with 4 × 30 screws and washers. They must not come out through the top: check the length against a scrap.

Check before moving on: Tight to both walls, and the slot sits over the channel.

3 · The oak uprights and shelves

Two uprights to the ceiling, the high shelf, the Series X shelf, the record ledge and the “now playing” ledge.

2–3 days · You + a helper · about £420–765

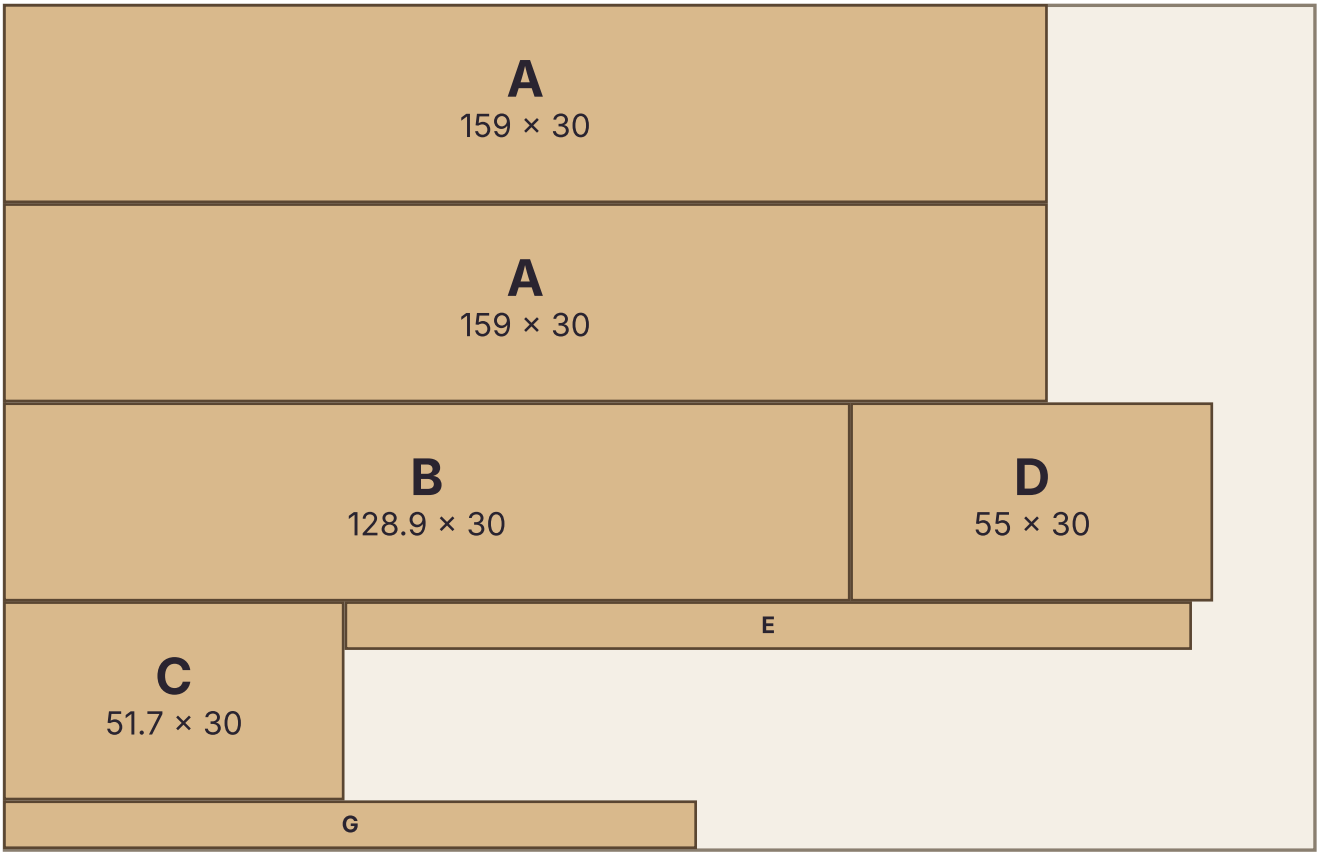


1 · Buy

Qty	What	Where	£
2	**Solid oak upright, 165 × 30 × 4.5** Two solid oak uprights, 165 × 30 × 4.5 cm (cut to 159), quarter-sawn if possible	Timber merchant	150–300
1 set	**Solid oak for the shelves and the side-wall run** Solid oak for the shelves: high shelf in two lengths (128.9 and 51.7 × 30 × 4), Series X shelf 55 × 30 × 4, record ledge 128.9 × 7 × 2.4, the side-wall run from one board 225 × 30 × 4 (cut to its sweep), lips	Timber merchant	230–380
1 set	**Oak wedges, dowel, concealed brackets** Oak wedges, 10 mm oak dowel for pegs and dowels, concealed upright brackets; two concealed shelf brackets for the PC shelf, rated 40 kg+ a pair	Timber merchant	30–60
1	**“Now playing” oak ledge and brackets** The “now playing” ledge on the breast, with the table by the stairs: oak 105.5 × 7 × 2.4 with a 1.5 lip (from the shelf board's offcut), two concealed brackets	Timber merchant	10–25
	Total		**420–765**

2 · Cut

Solid oak, hardwax oil: each piece to scale



Part	What	Size (cm)	Qty	From	Notes
A	Uprights	159 × 30 × 4.5	2	Solid oak	Buy 165 long and scribe to the ceiling; housings, mortises, LED and cable grooves cut on the bench
B	High shelf, long length	128.9 × 30 × 4	1	Solid oak	Housed 1.2 into both uprights and pegged
C	High shelf, short length	51.7 × 30 × 4	1	Solid oak	Housed into the left upright; cleat on the side wall
D	Series X shelf	55 × 30 × 4	1	Solid oak	Two through-tenons 6 × 2.4, wedged flush
E	Record ledge	128.9 × 7 × 2.4	1	Solid oak	Housed and pegged
F	Ledge lips	128.9 × 1.5 × 0.9 · 126 × 1.5 × 1.9	4	Solid oak	Glued along the ledge's and the run's front edges
G	“Now playing” ledge	105.5 × 7 × 2.4, lip 1.5	1	Solid oak	On the breast at 104, two concealed brackets
H	Wedges · pegs · dowels	6 × 3, tapering · 10 mm dowel, 5 long	2 + 5 + 8	Solid oak	Glued

Sizes are finished sizes, length × width × thickness; a 4 mm saw cut between pieces. Ply parts can be cut by the merchant from these layouts.

3 · Build

Cut the uprights to length

1. Measure from the worktop to the ceiling at each upright's position. The plan says 159; ceilings vary.
2. Cut each upright 2 mm under its measurement, then scribe its top to the ceiling as you did the worktop.
3. Mark each upright's faces: which side faces the TV, and the back edge that goes against the wall.

The housings (both uprights)

4. Measure from each upright's foot (the end on the worktop). The high shelf's housing is 116–120 up, the full 30 depth, 1.2 deep, on the TV-side face of both uprights.
5. The record ledge's housing is 77–79.4 up, 0–7 from the back edge, 1.2 deep, on the TV-side face of both.
6. On the left upright's other face (the turntable side), a second high shelf housing at 116–120, full depth, for the short length.
7. Clamp a straight batten as a guide, set the router 4 mm deep and rout, then step down to 12 mm in passes. Square the ends with a chisel. Practise on an offcut first.

The through-mortises (left upright)

8. Two mortises, each 6 wide and 2.4 tall, at 54.8–57.2 up from the foot, 12–18 and 20–26 cm from the back edge.
9. Mark them on both faces with a square and a knife. Drill most of each out with the 20 mm bit, halfway from each face, then chisel to the lines.
10. On the TV-side face, flare the top and bottom of each mortise outwards 3–4 mm, for the wedges.

The tenons (Series X shelf)

11. On the shelf's west end (the end that goes through the upright), mark two tenons 6 wide and 2.4 thick, centred in its 4 cm thickness, 12–18 and 20–26 from its back edge, 4.5 long.
12. Saw the shoulders and cheeks, and cut away the rest of the end back to the shoulder line.
13. Saw a kerf 3 cm deep across the middle of each tenon, parallel to the shelf's faces.
14. Test-fit through the mortises: snug, pushed in by hand.

The grooves

15. On each upright's TV-side face, rout a groove for the LED channel (usually about 1.7 wide × 0.8 deep: check yours), 2 cm in from the front edge, from 2 cm above the worktop to just under the high shelf's housing.
16. On each upright's back edge, rout a 2 × 1.5 cable groove the full length: the left one for the Series X's HDMI and power and the vinyl lead, the right one for Deco B's 12 V and patch leads.
17. Notch both uprights' tops for the ceiling LED strip.

Dry-fit, then fix

18. Lay each upright on the floor with its shelves in their housings. Check every joint closes before any glue.
19. Drill two 10 mm holes 3 cm deep in each upright's foot, and matching holes in the worktop at the marked positions (dowel centres make this exact). Glue the dowels into the uprights only, so they can come out.
20. Lay the Series X's and the vinyl leads into the left upright's back groove, and Deco B's into the right one's.
21. Stand the right upright on its dowels against the breast's side. Screw it to the reveal through three countersunk holes, with plugs, and plug the holes with oak.
22. Stand the left upright on its dowels, check it's plumb with the laser, and fix it to the back wall with the concealed brackets.
23. Slide the long high shelf and the record ledge into their housings from the front.
24. Screw a 30 cm cleat to the breast's side face (the reveal) for the short high shelf, and another at the Series X shelf's underside height. Slide the short high shelf in from the front onto its cleat and into its housing.
25. Feed the Series X shelf's tenons through the mortises from the turntable side, onto its cleat. Glue the wedges and drive them into the kerfs from the TV side. When dry, saw them flush and plane: proud, they'd touch the TV.
26. Pegs: drill 10 mm through each upright's face into the housed shelf ends, 4 cm deep, about 4 cm from the front edge. Glue in a peg and saw it flush.

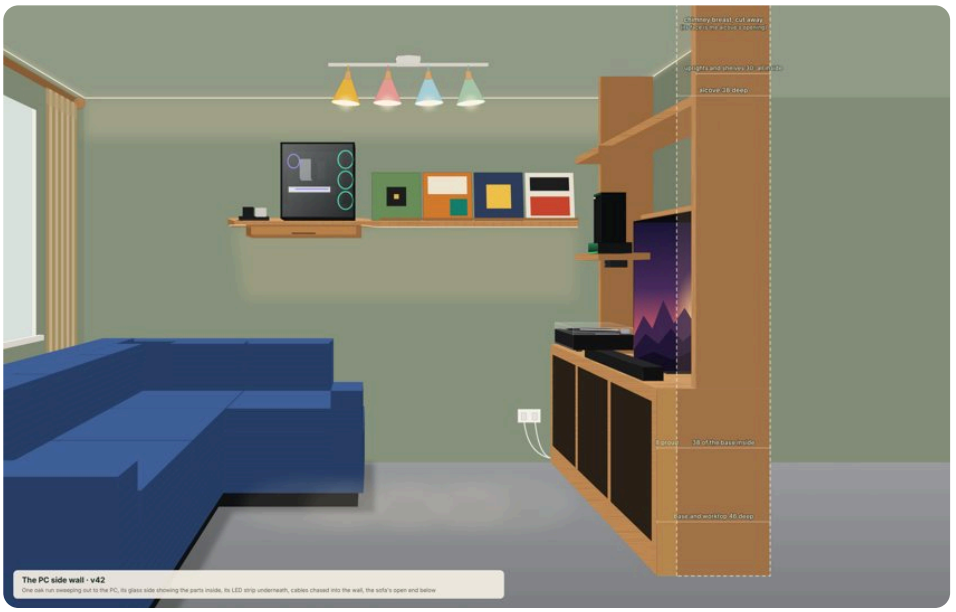
The “now playing” ledge on the breast

27. Cut it 105.5 × 7 × 2.4 from the shelf board's offcut and glue a 1.5 cm lip along its front edge, so a sleeve stands against the wall without sliding.
28. Mark a level line on the chimney breast at 101.6 (the ledge's underside), 6 cm in from each side of the breast.
29. Fit its two concealed brackets to the maker's instructions, slide the ledge on, and stand three LPs on it.

4 · The side-wall run and the PC shelf

One oak run from the corner to the PC, the PC shelf and its keyboard tray.

1 day · You · about £60–130

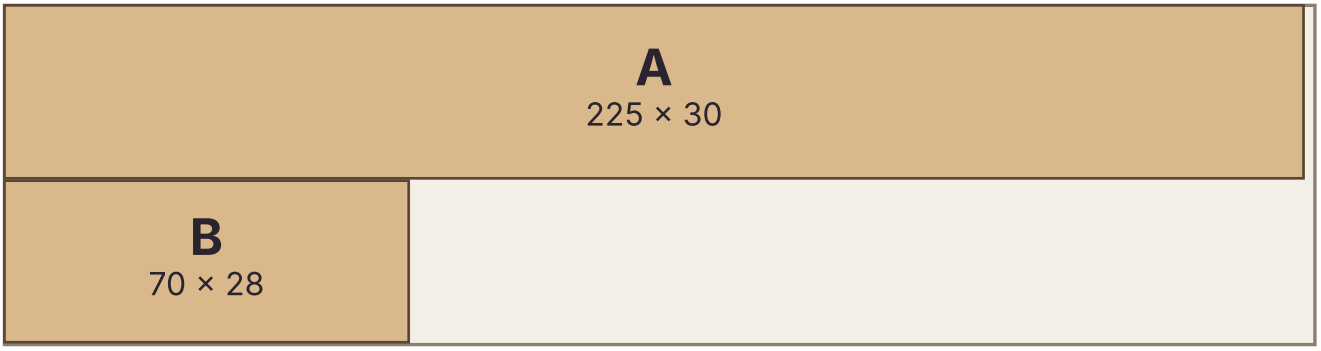


1 · Buy

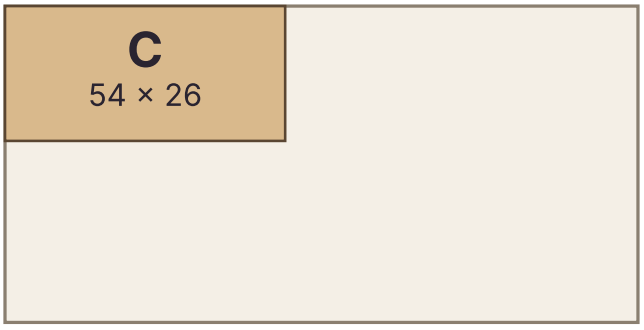
Qty	What	Where	£
1	**PC tidy kit: sleeve, grommet** PC tidy kit: braided sleeve, shelf-notch grommet	DIY shop or online	5–15
1	**Powered 4-port USB 3 hub** A powered USB 3 hub, 4-port with its own adapter, screwed under the side run's front edge below the PC: pads, the keyboard and a stick within reach of the sofa	DIY shop or online	15–30
1	**Remote PC power button** A remote power button for the PC: a wireless switch on the motherboard's power header (card in a spare slot), its push button on the wrap ledge	DIY shop or online	10–30
1	**Magnetic dust filter** A magnetic mesh dust filter cut to the PC's top exhaust	DIY shop or online	5–10
1	**Keyboard tray: oak housing, ply tray, runners** Keyboard tray for the PC shelf: 7 cm oak housing under it, 12 mm ply tray 54 × 26 with an oak lip, 300 mm light-duty runners	Timber merchant	25–45
	Total		**60–130**

2 · Cut

Solid oak, hardwax oil: each piece to scale



Birch ply, 12 mm: board 1 of 1, 122 × 61



Part	What	Size (cm)	Qty	From	Notes
A	Side-wall run	220 long × 7 sweeping to 28 × 4, from one 225 × 30 × 4 board	1	Solid oak	Open at back, seat, concealed brackets under the PC end; LED groove under the front edge
B	Keyboard tray housing	70 × 28 × 7, sides 7 × 2	1	Solid oak	Screwed under the PC shelf
C	Keyboard tray	54 × 26	1	Birch ply	Oak lip on its front edge

Sizes are finished sizes, length × width × thickness; a 4 mm saw cut between pieces. Ply parts can be cut by the merchant from these layouts.

3 · Build

Tools

- SDS drill
- Drill-driver
- Dowel jig
- Spirit level and laser
- Jigsaw
- Clamps

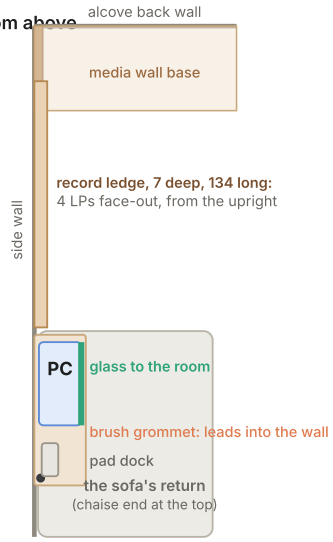
Fixings

- Two concealed shelf brackets rated 40 kg+ a pair
- 300 mm light-duty runners, one pair
- An oak-coloured brush grommet
- 10 mm dowels (2)
- 5 × 60 mm screws and plugs

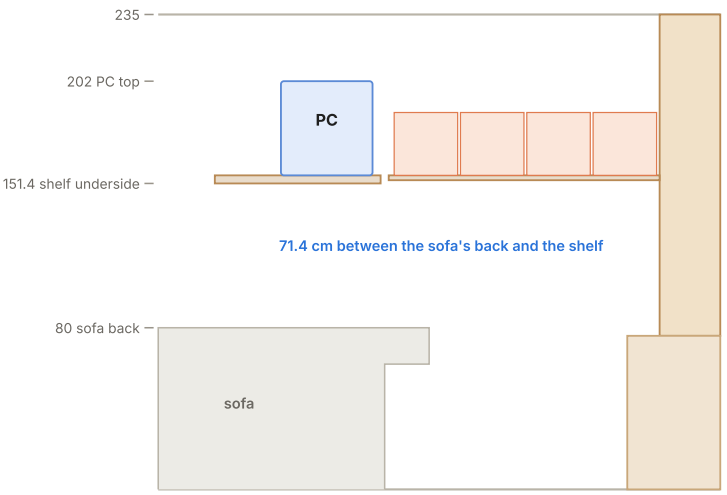
The side wall: the ledge wraps the corner, the PC sits above the sofa

One oak line at 155.4 cm: the record ledge from the upright, then the PC shelf over the end of the sofa, clear of the TV's sightlines

From above



From the room, looking at the side wall



The cleats

1. Screw the first cleat to the side wall with its top on the 153 line, from 30 cm out from the back wall (in line with the uprights' fronts) to 164 cm out. Plugs every 25 cm.
2. Screw the second with its top at 151.4 (the PC shelf's underside), from 168 to 250 cm out.

The wrap ledge

3. Lay the ledge on its cleat, its alcove end in line with the uprights' fronts, and screw up through the cleat into it.
4. Glue the lip along its front edge.

The PC shelf and its tray

5. Fit the concealed brackets into the wall at the shelf's height, through the cleat, to the maker's instructions.
6. Cut the grommet hole in the shelf's back corner, over the top of the chased conduit.
7. Screw the keyboard tray's housing to the shelf's underside, stopping 12 cm short of the sofa end so the cables drop past it. Fit the runners inside it and slide the tray in.
8. Slide the shelf onto its brackets and down onto the cleat. Lean on it: it shouldn't move.

The PC's leads

9. Fit the brush grommet. Pull the HDMI (active optical: the end marked "source" at the PC) and the Cat6 down the 38 mm conduit with its draw wire.
10. Take the plug off the C13 lead, feed its bare end down the 25 mm conduit, and at the bottom fit the rewireable plug: brown to L, blue to N, green-and-yellow to E, the cord grip on the outer sheath, a 13 A fuse.
11. At the bottom, all three run behind the skirting into the plinth's notch and the channel; the mains lead goes on to the 10-gang behind the TV.

The PC's extras

12. Screw the powered USB hub under the side run's front edge, below the PC, ports facing the sofa. Its adapter plugs into the 10-gang along the PC's mains route; its lead runs up through the grommet to the PC.
13. Remote power button: with the PC unplugged, push the receiver card into a spare slot and plug its lead onto the motherboard's power-switch pins (the case's own button lead plugs into the card's pass-through). Stick the

push button to the wrap ledge's underside, by the sofa end.

14. Lay the magnetic dust filter on the PC's top exhaust; brush it off once a month.

Check before moving on: One level line with the ledge over the TV, the sweep runs smooth with no flat spots, and the PC end holds your weight leaning on it.

5 · The dining cabinet and table

Two oak boxes under your shelf: the table folds into one, the six chairs live in the other.

3–4 days · You + a helper · about £906–1,535

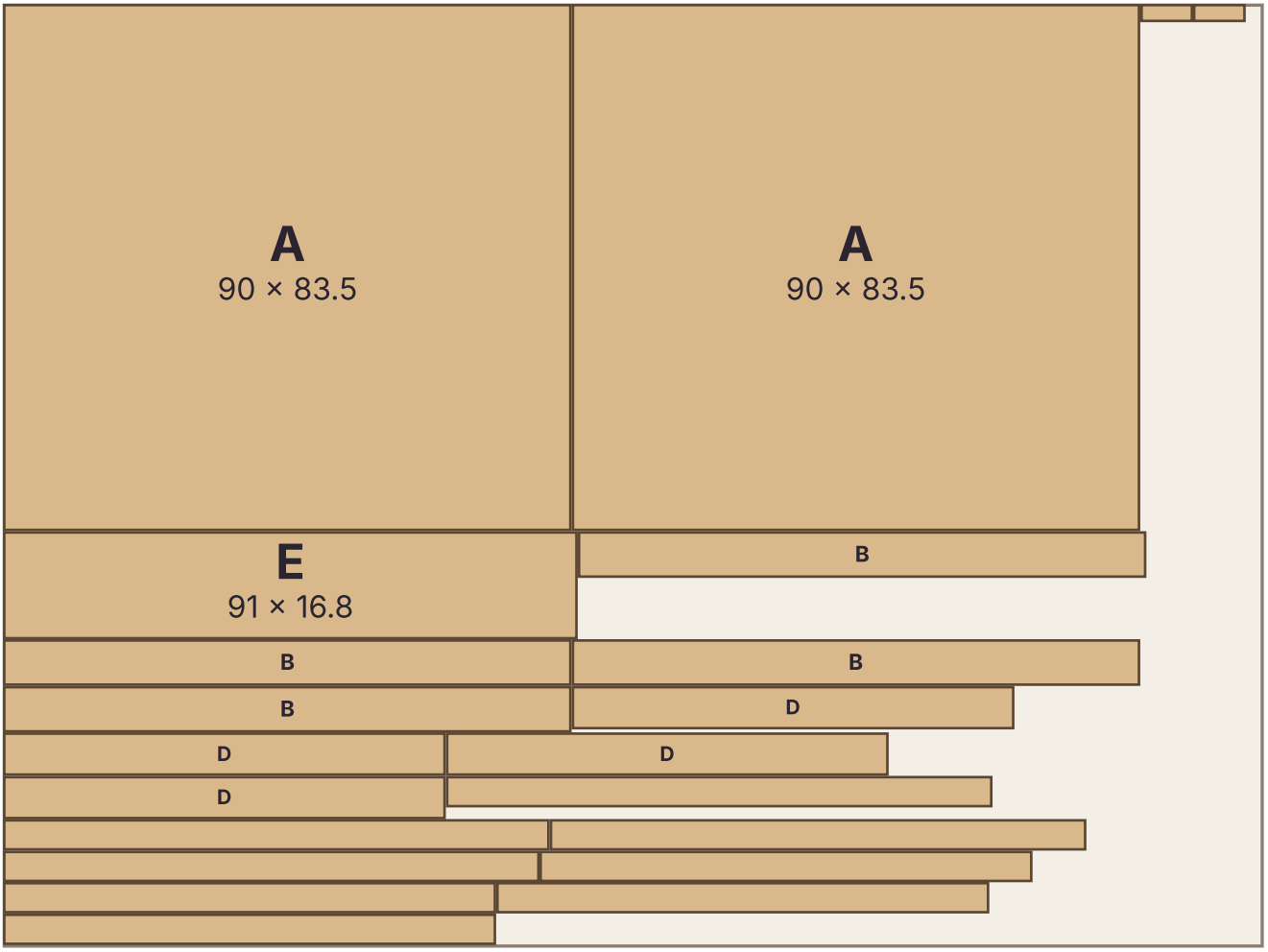


1 · Buy

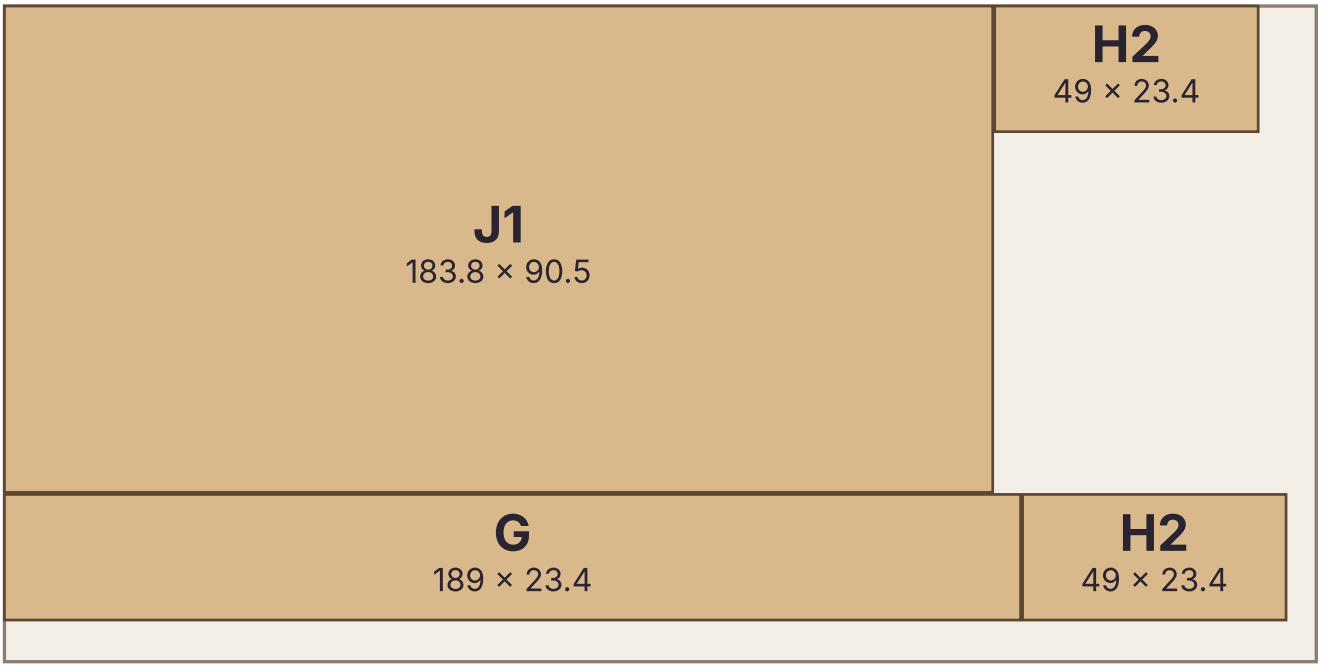
Qty	What	Where	£
1 set	**Solid oak for the table, two 92.5 × 90 halves** Solid oak for the table: two halves 90 × 92.5 × 2.7, each a centre 78.5 × 90 of edge-glued staves with two pegged breadboard ends (the cutting list)	Timber merchant	265–460
1 set	**Oak-faced ply, cut to size: two boxes, two doors** The dining cabinet: oak-faced ply cut to size, 18 mm for its two boxes (four ends, two tops, two floors), the linen shelf, the chair shelf and the two doors (90.5 and 45.5 × 183.8); a 6 mm ply back for each box (the cutting list)	Timber merchant	230–410
1 set	**Oak: hinge rail, aprons, four legs, door stops** Oak for the cabinet's hinge rail (91 × 15.4 × 7), the eight apron rails, the four tapered legs (6.5 × 4.5) and two door stops	Timber merchant	35–70
1	**Piano hinge, 1800 × 50 (the table)** Altro continuous piano hinge, 1800 × 50, cut to 90: the table's foot	Online (Altro)	16–20
1 set	**Folding leg brackets (4), tabletop fasteners (16)** Folding, self-locking table-leg brackets, a set of four, and sixteen figure-8 tabletop fasteners for the aprons	DIY shop or online	60–75
1 set	**Two door piano hinges, catches, magnet, hooks** The doors' hinges and catches: two Altro continuous piano hinges, 1800 × 32 × 1 mm, stainless, one the height of each door; four magnetic catches (the doors and the folded table), a 10 mm flush magnet and two steel cups to hold an open door flat, a small brass hook for the cloth bag, two steel angle brackets for the wall	Online (Altro)	35–65
3 pairs	**Grey padded folding chairs (Brayden Studio)** Folding chairs, six: your grey padded ones (Brayden Studio on Wayfair, sold in pairs, so three sets of two): wooden frames, grey linen, a 47 × 47 seat with a 6.5 cm cushion, a 110° back, 168 kg; folded 90 × 10 × 47, they live in the cabinet. The estimate is about £80–130 a pair: read the listing's price	Wayfair	240–390
3	**Lee Valley flip-top hinge** Three Lee Valley 40574 narrow flip-top hinges (solid brass, 1.27 × 7.0 cm) for the joint	Lee Valley	25–45
	Total		**906–1,535**

2 · Cut

Solid oak, hardwax oil: each piece to scale



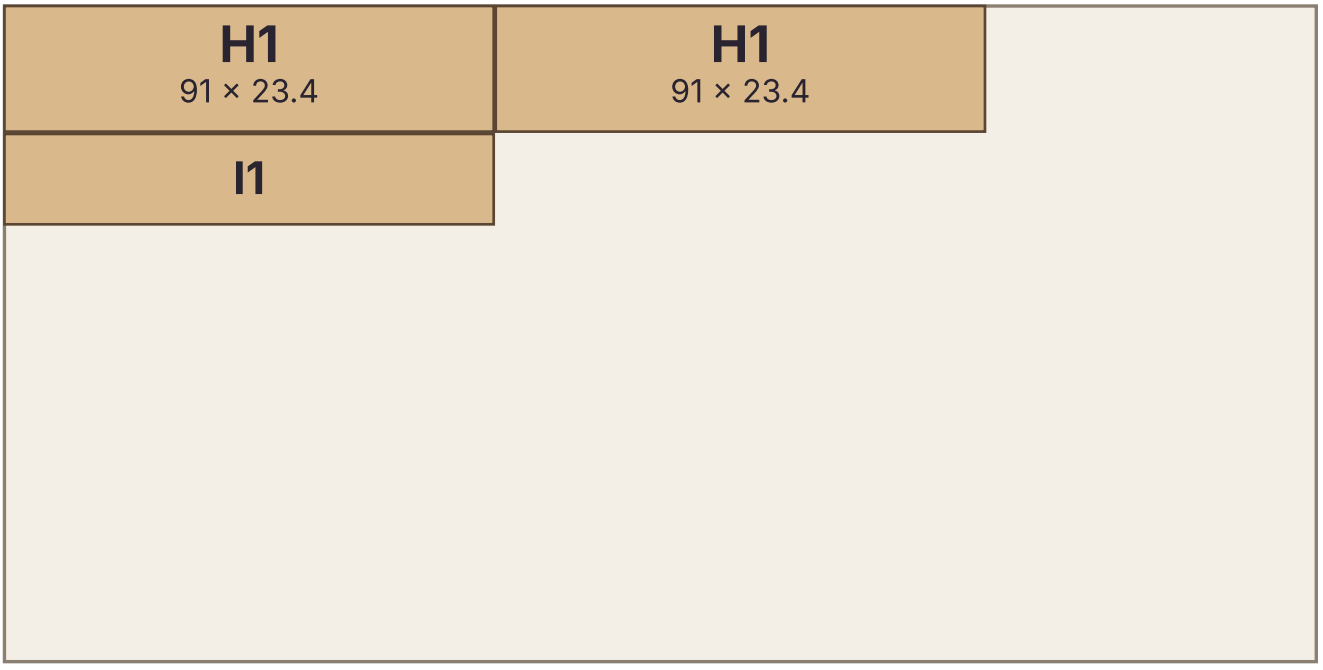
Oak-faced ply, 18 mm: board 1 of 3, 244 × 122



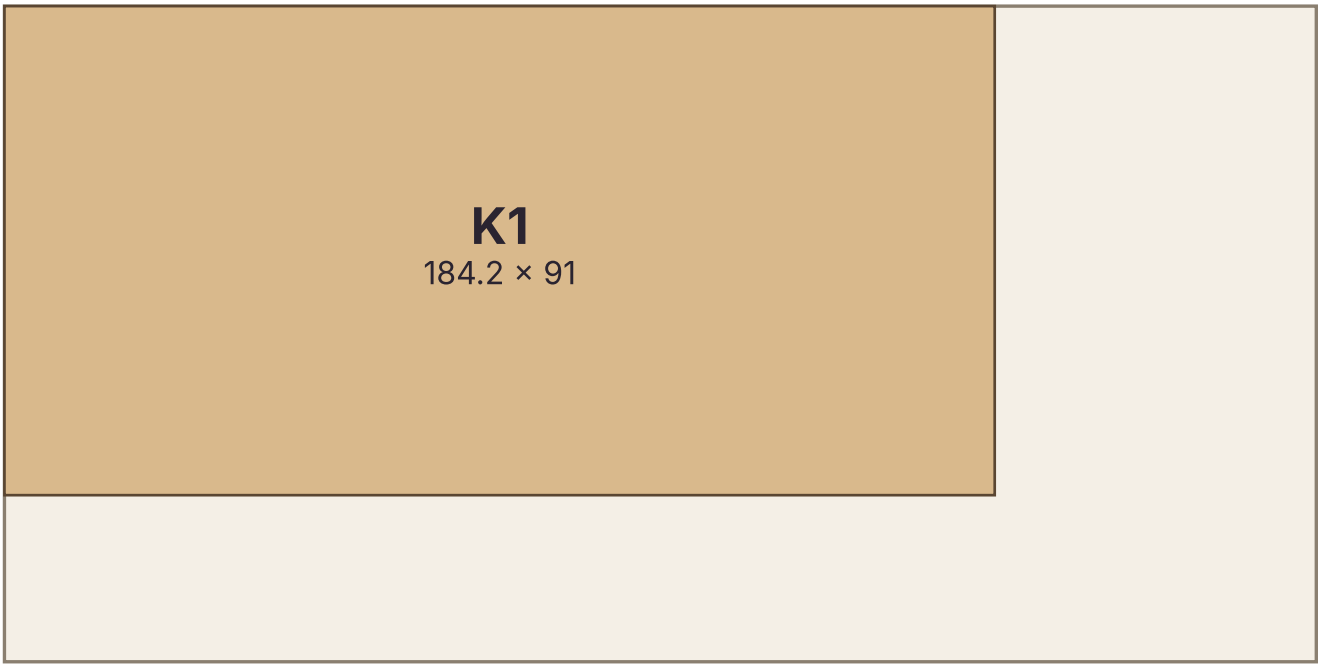
Oak-faced ply, 18 mm: board 2 of 3, 244 × 122



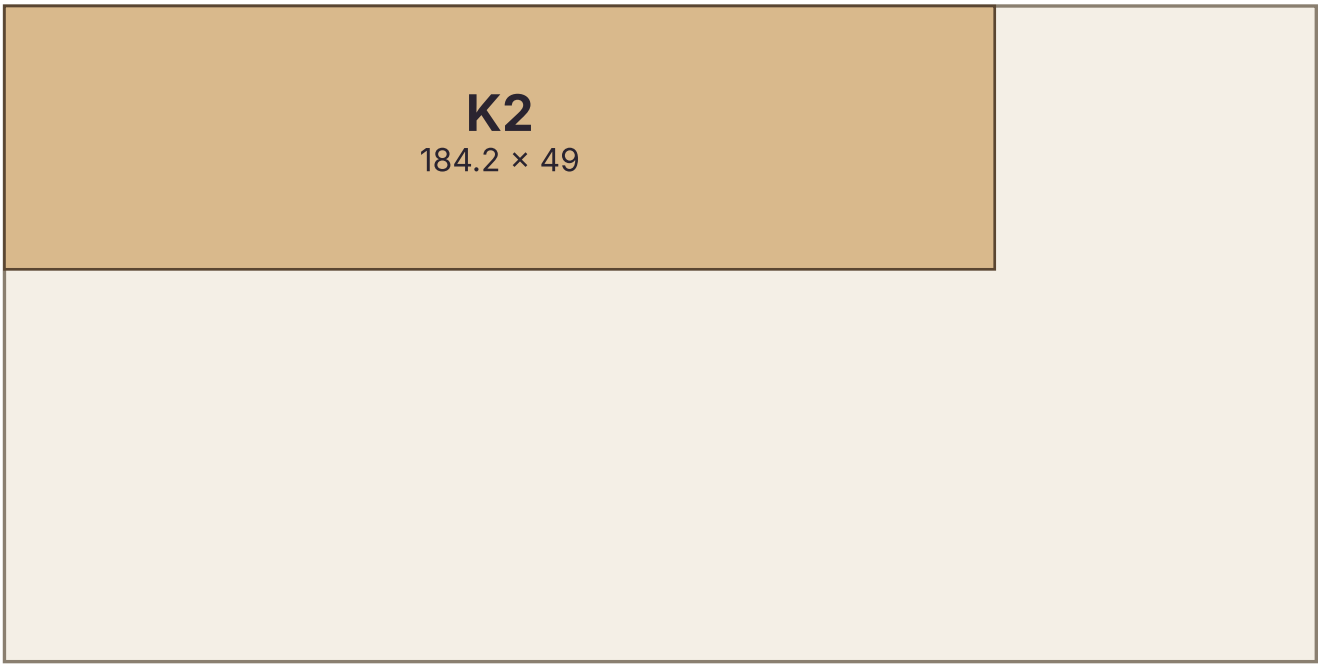
Oak-faced ply, 18 mm: board 3 of 3, 244 × 122



Birch ply, 6 mm: board 1 of 2, 244 × 122



Birch ply, 6 mm: board 2 of 2, 244 × 122



Part	What	Size (cm)	Qty	From	Notes
A	Table halves, centres	83.5 × 90 × 2.7	2	Solid oak	Edge-glued staves: 78.5 between the shoulders, plus a 2.5 tongue each end
B	Breadboard ends	90 × 7 × 2.7	4	Solid oak	Pegged with three 10 mm pegs each, in slotted holes
C	Apron rails	outer half 86.5 and 78 · inner half 82.5 and 78, all × 4.6 × 2	2 halves	Solid oak	A frame under each half, held on figure-8 fasteners
D	Table legs	6.5 × 4.5, cut 70 long	4	Solid oak	Tapered on the two inside faces from 25 cm down to 4.5 × 3.5 at the foot; trimmed to the floor once the table's hung and level
E	Hinge rail	91 × 16.8 × 7	1	Solid oak	Across the table box, its top at 73.3, its front 4 behind the door's back (the cloth bag hangs past it); the table's piano hinge 4.6 back from its front

F	Door stops (the cabinet)	8 × 2.5 × 1.2	2	Solid oak	Under each box's top, behind its door's free edge; a magnetic catch let into each
G	Cabinet ends	189 × 23.4	4	Oak-faced ply	Full height and depth, two for each box; the doors sit between them
H	Cabinet tops · floors	91 × 23.4 · 49 × 23.4	2 · 2	Oak-faced ply	One of each per box, between its ends; the floors' top faces at 3
I	Cabinet linen shelf · chair shelf	91 × 16.8 · 49 × 20.8	1 · 1	Oak-faced ply	Top faces at 168.6 (the table box, set 4 back for the cloth bag) and 95.8 (the chair box)
J	Cabinet doors	90.5 · 48.5, both × 183.8	1 · 1	Oak-faced ply	The table door, then the chair door: oak-faced both sides, grain upright, all four edges lipped; a piano hinge on the edge by the boxes' join, a finger slot near the other
K	Cabinet backs	91 · 49, both × 184.2	1 · 1	Birch ply	Glued and pinned in each box, between its ends, its floor and its top

Sizes are finished sizes, length × width × thickness; a 4 mm saw cut between pieces. Ply parts can be cut by the merchant from these layouts.

3 · Build

Tools

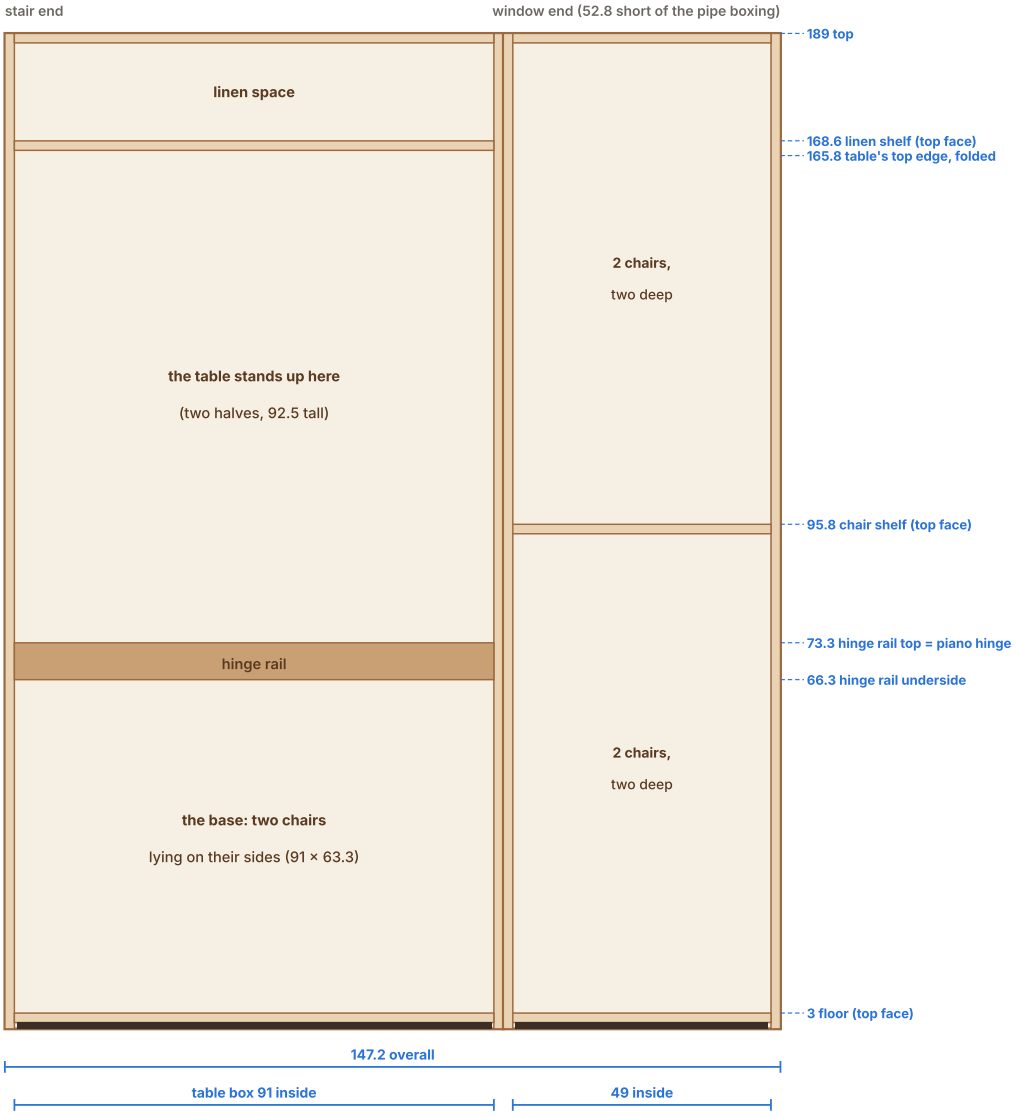
- Drill-driver and a countersink
- A self-centring (Vix) bit for the piano hinges' pilot holes
- 35 mm and 10 mm Forstner bits
- Router with straight bits and a guide
- Jigsaw or track saw
- Hacksaw (to cut the table's piano hinge)
- Chisels
- Clamps, at least four 100 cm
- Long spirit level and a square
- Tape measure and pencil
- A helper for standing the boxes and hanging the table

Fixings

- Altro continuous piano hinge, 1800 × 50, cut to 90 (the table's foot)
- Two Altro continuous piano hinges, 1800 × 32 × 1 mm, stainless (the doors), not cut
- Three Lee Valley 40574 narrow flip-top hinges (1.27 × 7.0) and their screws
- Folding, self-locking table-leg brackets, a set of four
- Sixteen figure-8 tabletop fasteners
- Four magnetic catches
- A 10 mm flush magnet and two steel cups
- A small brass cup hook (the cloth bag)
- Two steel angle brackets, plugs and 5 × 70 mm screws for your solid wall
- 4 × 50, 4 × 30 and 3.5 × 16 mm wood screws (the short ones for the piano hinges), 25 mm panel pins, wood glue
- Plastic shims, felt pads, hardwax oil, 10 mm oak dowel for the breadboard pegs

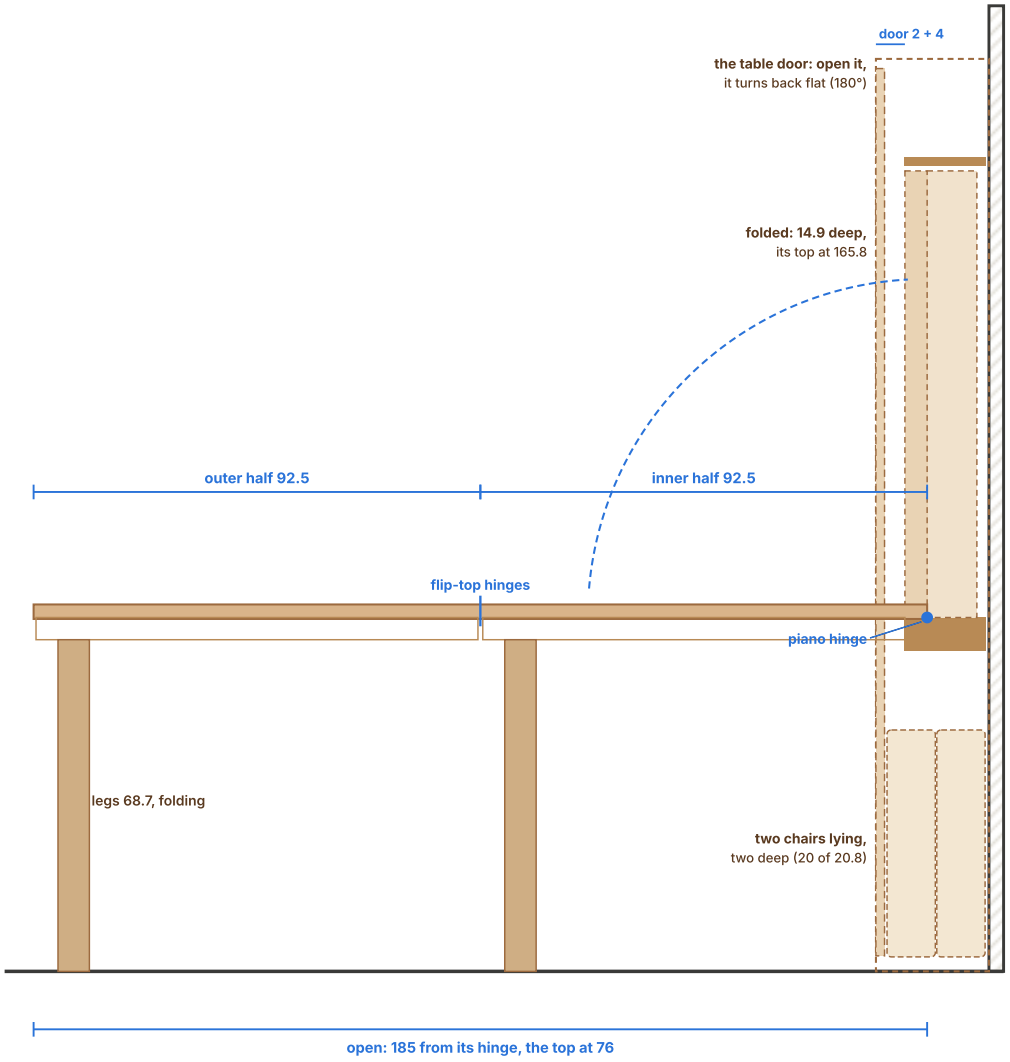
The compact cabinet's two boxes, from the front

Doors and table off; heights to each panel's top face. Ends, tops and floors 23.4 deep; inside 20.8 deep, 2 back for the inset doors.



The compact cabinet: how the table comes down

Through the table box, wall on the right. Folded (dashed): 4 back from the door, two chairs lying under it. Open: the door turned back, the table down.



Before you start: check and mark out

1. Measure the gap under your shelf at both ends: it must be at least 191. The boxes are 189 tall and need 2 cm to tip up into place.
2. Mark the cabinet's footprint on the floor: 147.2 long and 23.4 deep off the wall, its stair end 45 cm short of the stair box (that strip stays clear for the stairs' foot). Its window end is then 52.8 cm short of the pipe boxing's face. Mark the line between the two boxes, 94.6 from the stair end.
3. Your stairs wall is solid: the two brackets go in with plugs and screws. Run the detector over both spots first, as for every hole.
4. Lay a long level along the footprint. Note the low end: the shims go there later.
5. Oil every oak face and edge (two thin coats, a day apart) before you assemble anything: it's much easier on flat parts, and a board oiled on one side cups.

The two boxes

6. Each box is two ends with a top and a floor between them, a back and a shelf. Lay the ends flat, inside faces up, and mark on each: the floor's top face at 3, the top flush with the end's top edge, and a line 2 back from the front edge, full height. The shelves and the hinge rail stand behind that line; the door sits in front of it, flush with the front edges.
7. On the table box's two ends mark the hinge rail (66.3–73.3) and the linen shelf's top face at 168.6. On the chair box's ends mark the chair shelf's top face at 95.8.

8. Stand two ends up and fit their top and floor between them, flush at the front: glue, pre-drill and drive three 4 × 50 screws through each end into each panel's end grain. The table box is 91 inside, the chair box 49. Each floor's bottom face is at 1.2.
9. Fit the chair shelf in the chair box, its front edge on the line, and the linen shelf in the table box 4 behind the line (a hook for the cloth bag goes in front of it): glue and three screws through each end.
10. Measure both diagonals of each box's front: when they're equal, it's square. Glue and pin each 6 mm back (91 or 49 × 184.2) between the ends, the floor and the top; it holds the box square.
11. Glue and screw two battens under each floor, one 1 cm back from the front edge and one at the back: that's the plinth.

Stand them and fix them to the wall

12. With your helper, tip the table box up and walk it onto its marks, back to the wall; then the chair box beside it at the window end.
13. Shim under the battens until both are level along their length, plumb at the front, and flush with each other at the front and the top. Check again after a minute: they settle.
14. Clamp the two meeting ends together and screw them to each other with four 4 × 30 screws from inside the chair box, at about 20, 70, 120 and 170 up. Now they're one cabinet.
15. Screw an angle bracket under each box's top, in the middle, and into the wall with plugs and 5 × 70 screws. They stop it tipping; the floor takes the weight.

The hinge rail

16. Slide the oak hinge rail (91 × 16.8 × 7) into the table box against the back, its top face at 73.3: its front stands 4 behind the line, like the folded table's, so the cloth bag on the door's hook clears it.
17. Glue it, then drive two 4 × 50 screws through each end into it. It carries the table's whole weight when it's open, so don't skip the glue.

The two table halves

18. Ask the timber merchant for two centre panels of edge-glued staves, each 83.5 × 90 × 2.7: 78.5 between the shoulders plus a 2.5 tongue at each end, and four breadboard ends 90 × 7 × 2.7.
19. On each end of each centre panel, rout a tongue 2.5 long and 0.9 thick, centred in its thickness: take 0.9 off each face with the router against a batten.
20. In each breadboard end's inner edge, rout a groove 0.9 wide and 2.6 deep, centred, full length. Dry-fit: flush with the panel's long edges and flat across the joint.
21. Drill three 10 mm peg holes down through each breadboard end and its tongue: one at the centre and one 8 cm in from each side, 1.3 cm from the breadboard's inner edge. Take the ends off and elongate the two outer holes in each tongue to 16 mm along the width, so the panel can move.
22. Glue only the middle 10 cm of each tongue, clamp the ends on and drive the pegs (glue the outer pegs into the breadboard only). Saw flush, sand to 180 and oil both faces equally.
23. Each half is now 92.5 × 90. Chamfer the underside edges 1 cm, except the two edges that will meet at the joint.

The aprons (the frame under each half)

24. Lay both halves face down on a clean blanket, their joint edges together, as on the drawing of the halves from underneath.
25. The outer half: its two long rails (86.5) run along the half 4 cm in from the sides and 3 cm in from both ends; its two short rails (78) fit between them at the ends. Glue and screw the corners (two 4 × 50 each).
26. The inner half: the same, but its frame stops 4.6 short of the wall end (that strip rests on the hinge rail): long rails 84.9, short rails 78.
27. Fix each frame to its half with figure-8 tabletop fasteners every 25 cm (four a rail): recess each in the rail's top edge with the 35 mm bit, screw it to the rail and then up into the top with 4 × 30 screws. They hold the top flat

but let it move.

The legs

28. Cut the four legs 70 long from the 6.5 × 4.5 oak. They're trimmed to the floor at the end.
29. Screw a folding bracket to the underside of each half where its legs stand (see the drawing): on the inner half against the end rail by the joint, the leg centres 8.25 cm from the joint and 32 cm in from each side; on the outer half in the two corners at the far end, centres 8.25 cm in from the end and the sides.
30. Screw each leg's top into its bracket. Fold them: the inner half's legs fold towards the wall end, the outer half's towards the joint. Folded, each leg lies flat inside its apron frame.

Join the halves: three flip-top hinges

31. Turn the halves face up, their meeting edges together, flush across the top. Mark three hinges along the joint: 10 cm in from each side and one in the middle.
32. Rout each hinge's two leaves into the two edges to the hinge's depth with a guide, square the recesses with a chisel and screw the hinges in.
33. Swing it: open, the two tops are flush; folded, the outer half lies flat on the inner half, top face to top face, with nothing proud.

Hang the table

34. Cut the 50 mm piano hinge to 90 with a hacksaw and file the ends smooth.
35. With the inner half face down, screw one leaf of the piano hinge along its wall end, on the underside, the knuckle exactly on the end edge: a screw every 10 cm, pilot holes first.
36. Fold the outer half onto the inner half. With your helper, lift the folded table upright into the table box, its wall end down on the hinge rail, its underside facing the room, centred (0.5 each side).
37. Screw the hinge's other leaf to the rail's top face, the knuckle 4.6 back from the rail's front. That leaves 4 in front of the folded table for the cloth bag on the door. One screw at each end first, try the swing, then the rest.
38. Try it: it swings down to flat with its top at 76 and back up to stand in the box without touching the sides. The folded table's top edge is at 165.8, just under the linen shelf.

The catches and the finger pull

39. Screw two magnetic catches under the linen shelf's front edge, 15 cm in from each side, and their plates on the folded table's top edge: they hold it upright.
40. Rout a finger pull 12 long and 1.5 deep in the underside of the inner half's apron rail by the joint, in the middle: folded, it's at the top, and that's what you pull to bring the table down. No knob, so nothing presses into the pads on the door.

Hang the two doors

41. The table door (90.5 × 183.8) hangs on the table box's window end and the chair door (48.5 wide) on the chair box's stair end. The two hinges sit side by side where the boxes meet, and each door opens away from the other, turning 180° to lie flat over it.
42. Lay a door face down, its hinge edge up. Screw one leaf of a 32 mm piano hinge along that edge, centred (1.9 short of each end), the knuckle just proud of the door's face: a 3.5 × 16 screw every 10 cm into pilot holes made with the self-centring bit.
43. Stand the door in its opening on 2 mm packers, with an even gap all round. Screw the hinge's other leaf to the inside face of the end it hangs on, the knuckle just proud of the front edge: one screw at the top and one at the bottom first, try the swing, then the rest.
44. Hang the table door first, then the chair door. Each should shut flush with the front edges and open right back, flat against the other door.

45. Glue an oak door stop under each box's top, behind its door's free edge, with a magnetic catch let into it and the catch's plate on the door's back: the door closes flush and stays shut.
46. Let the 10 mm flush magnet into the chair door's face, 36.1 from its hinge edge and 100 up, and the two steel cups into the table door's face at the same height, 32.2 and 40 from its hinge edge. Whichever door is open now holds flat against the other.
47. Screw a small brass hook to the table door's back, high up in the middle: the cloth bag (the tablecloth and napkins) hangs on it, in the space in front of the folded table.
48. Rout an upright finger slot in each door 5 cm in from its free edge, 26 long, 2.5 wide and 1 deep, at about 92–118 from the floor; ease its edges and oil it.
49. Open the table door flat and press down gently on its far end: it mustn't drop. If it does, check every hinge screw is in tight before you load the cabinet.

Level the table and trim the legs

50. Open the table door and lay it back over the chair door. Open the table: pull it down, unfold the inner half's legs till they click, lower it; flip the outer half over and unfold its legs till they click.
51. Lay the level across it in both directions. Mark each leg where it should meet the floor so the top is level at 76, fold the legs, saw them off square and stick felt pads on.

Load the chairs and check

52. Lay two folded chairs on their sides in the table box's base, under the hinge rail, one behind the other: the back one first.
53. Open the chair door. Stand two folded chairs on the chair box's floor, one behind the other, and two on its shelf the same way.
54. Fold the table up, shut both doors and stand back: two plain oak doors, nothing on show.

Check before moving on: The table swings down and up without touching the box's sides, all four legs click, the top is level at 76 both ways, the catches hold it folded, and each door shuts flush with an even gap and turns back flat against the other without its far end dropping.

6 · The bar

A cupboard scribed between the cabinet and the wall, an oak counter, the ENHET frame and mirror over it.

1 day · You · about £120–200

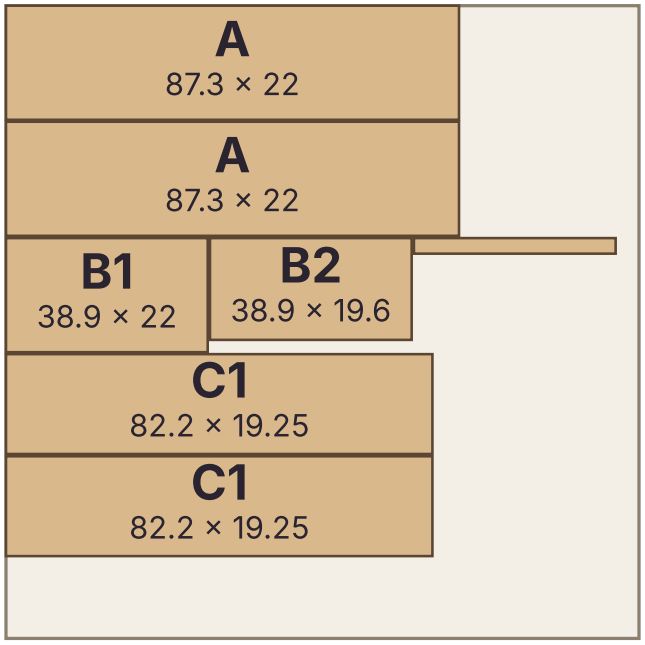


1 · Buy

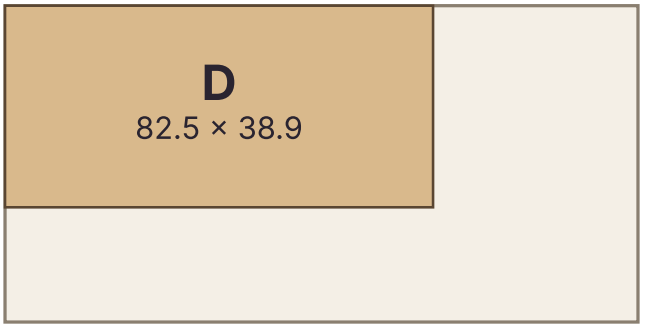
Qty	What	Where	£
1 set	**The bar: oak-faced ply cupboard, oak counter, hinges** The bar beside the dining cabinet: 18 mm oak-faced ply for a cupboard 42.5 × 22 × 87 on a 3 cm plinth (two ends, a floor, a shelf, a back, two doors), a 27 mm oak counter 42.5 × 22.4, two pairs of soft-close hinges, two brass knobs	Timber merchant	70–120
1 set	**Over the bar: IKEA ENHET frame, mirror, glass rail, LED** Over the bar: an IKEA ENHET wall frame with shelves, 40 × 15 × 75, anthracite, a 3 mm acrylic mirror cut to 42 × 97, a warm white LED strip	IKEA	50–80
	Total		**120–200**

2 · Cut

Oak-faced ply, 18 mm: board 1 of 1, 122



Birch ply, 6 mm: board 1 of 1, 122 × 61



Solid oak, hardwax oil: each piece to scale



Part	What	Size (cm)	Qty	From	Notes
A	Bar ends	87.3 × 22	2	Oak-faced ply	Scribed to the cabinet's end and the wall
B	Bar floor · shelf	38.9 × 22 · 38.9 × 19.6	1 · 1	Oak-faced ply	The floor on a 3 plinth set back 2; the shelf's top at 40.8
C	Bar doors · plinth front	82.2 × 19.25 · 38.9 × 3	2 · 1	Oak-faced ply	Soft-close hinges; edges lipped
D	Bar back	38.9 × 82.5	1	Birch ply	Glued and pinned between the ends
E	Bar counter	42.5 × 22.4 × 2.7	1	Solid oak	From the worktop's offcut

Sizes are finished sizes, length × width × thickness; a 4 mm saw cut between pieces. Ply parts can be cut by the merchant from these layouts.

3 · Build

The bar beside the cabinet

1. Measure the gap at the floor, at 45 and at 90: the cabinet's end to your shelf's end, less the 10 cm gap. Drawn: 42.5 wide, 22 deep.
2. Scribe the two ends to the cabinet's end and the wall, then glue and screw the floor, the shelf and the back between them; the plinth front set back 2.
3. Stand it, check it's level, and screw it to the wall through the back's top rail and to the cabinet's end (4 × 30, from inside).
4. Fix the oak counter on top, 2.7 thick, its front edge flush; hang the doors on soft-close hinges with a push catch (no handle to catch the table door).

5. Over it: the ENHET frame on the wall, its bottom at 112, plugged into the solid wall; the mirror behind it; the glasses on its shelves.

7 · The glass case

An oak frame on the chimney breast with three glass shelves and a lift-off glass front.

½ day · You · about £90–170

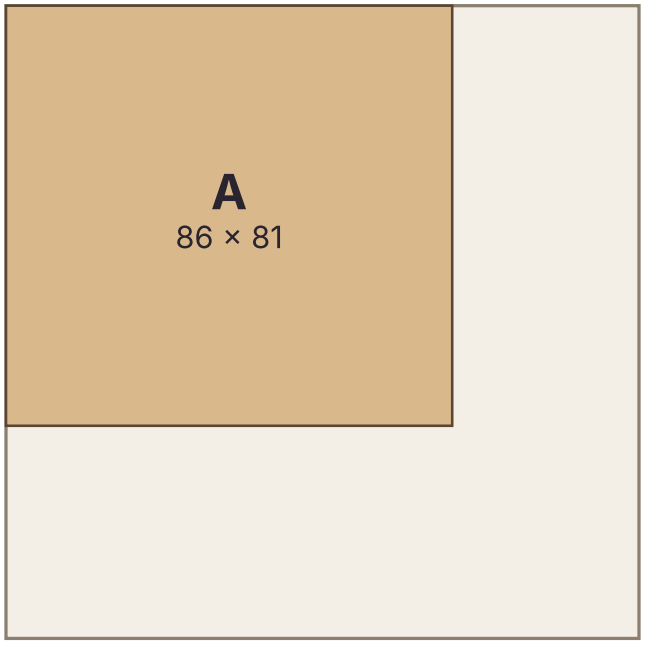


1 · Buy

Qty	What	Where	£
1 set	**Glass case: 4 mm toughened front, three 6 mm shelves, cut to size** The glass display case's glass: toughened, cut to size, polished edges: a 4 mm front 85.6 × 82.2 that lifts off, and three 6 mm shelves 85.6 × 9.2	A glass cutter	70–130
1 set	**Glass case: oak frame, oak-faced ply back, pins, cleat, LED strip** The glass display case's oak and fittings: the frame from the oak offcuts (two sides 85 × 12 × 2, top and bottom 86 × 12 × 2), a 6 mm oak-faced ply back 86 × 81, twelve shelf pins for glass, an oak French cleat 80 long, a warm white LED strip under the top on the strips' app	Timber merchant	20–40
	Total		**90–170**

2 · Cut

Oak-faced ply, 6 mm: board 1 of 1, 122 ×



Solid oak, hardwax oil: each piece to scale



Part	What	Size (cm)	Qty	From	Notes
A	Glass case back	86 × 81	1	Oak-faced ply	Pinned into the frame's rebate
B	Case sides	85 × 12 × 2	2	Solid oak	From the oak offcuts; rebated for the back
C	Case top and bottom	86 × 12 × 2	2	Solid oak	Between the sides

Sizes are finished sizes, length × width × thickness; a 4 mm saw cut between pieces. Ply parts can be cut by the merchant from these layouts.

3 · Build

The glass display case on the breast

1. Rout a 5 mm groove 6 mm deep along the inside of the top and bottom pieces, 1.5 cm in from their front edges, for the glass front: the top's groove 12 mm deep, so the front lifts up into it and drops into the bottom one.
2. Rout a 6 mm rebate along the back edges of all four pieces for the ply back.
3. Drill the shelf-pin holes on the sides' inner faces, front and back, 3 cm in from each edge: at 19, 40 and 61 above the bottom piece's top (the shelves at 158, 179 and 200 on the wall).
4. Glue and dowel the frame together (sides between the top and the bottom), check the diagonals, then glue and pin the ply back into its rebate.
5. Stick the LED strip under the top, just behind the groove, and run its lead out through a 10 mm hole in the back's top corner, down the breast's side to a plug (the strips' app runs it).
6. Screw one cleat strip to the case's back, 10 under its top, the bevel down and facing the wall. Screw the other to the breast at 212 (its top edge), level, the bevel up and facing out, with 5 × 60 screws in plugs every 20 cm.

7. Lift the case onto the cleat (two people: about 16 kg before the glass), check it's level, and fix its bottom edge to the breast with two small mirror plates.
8. Fit the pins and lay in the three glass shelves; lift the front up into the top groove and drop it into the bottom one. Fill it.

8 · The record table

Two KALLAX on their backs on a ply base with castors, under a hinged oak lid.

½ day · You · about £170–210

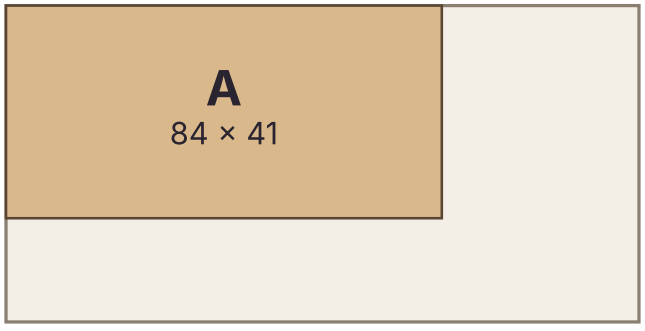


1 · Buy

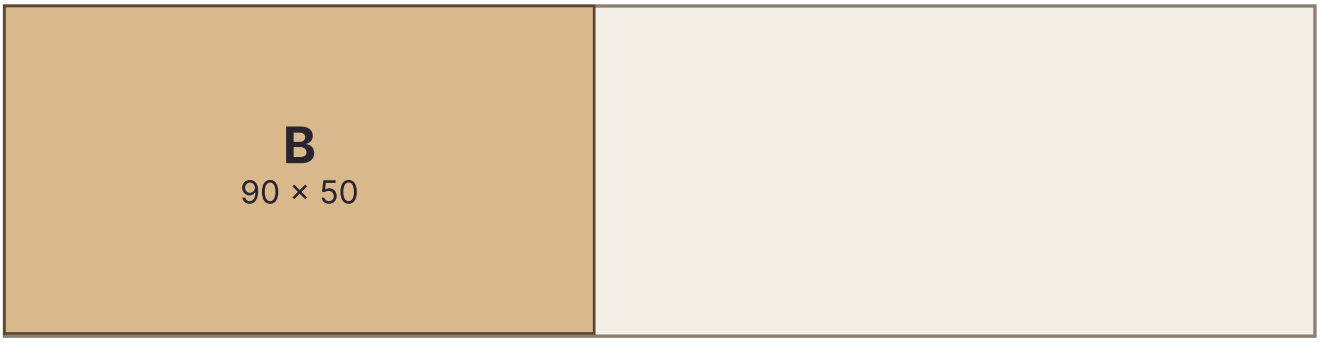
Qty	What	Where	£
1 set	**Record table: two KALLAX, oak lid, ply base, castors** The record table: two IKEA KALLAX 42 × 42, an 18 mm oak board 60 × 120 (the lid, 90 × 50), 12 mm ply for the base, four 50 mm castors, a piano hinge cut to 83, two lid stays	IKEA	170–210
	Total		**170–210**

2 · Cut

Birch ply, 12 mm: board 1 of 1, 122 × 61



Solid oak, hardwax oil: each piece to scale



Part	What	Size (cm)	Qty	From	Notes
A	Record table base	84 × 41	1	Birch ply	Screwed to the two KALLAX's open backs; the castors under it
B	Record table lid	90 × 50 × 1.8	1	Solid oak	From one 18 mm oak board 60 × 120; piano hinge along its back

Sizes are finished sizes, length × width × thickness; a 4 mm saw cut between pieces. Ply parts can be cut by the merchant from these layouts.

3 · Build

The record table

1. Assemble the two KALLAX as IKEA shows, then lay them on their backs side by side and screw them together through the sides (4 × 30).
2. Screw the 12 mm ply base to their open backs and the four castors to the base, 3 cm in from each corner.
3. Hinge the oak lid (90 × 50) along one long side; fit both lid stays so it can't drop. Full of LPs it's about 65 kg: roll it, don't lift it.

9 · Bought and fitted: shelves, cabinets, the door shelf

The alcove shelves, the two wall cabinets and the shelf over the door with its boxes. Nothing to cut but the door shelf.

1 day · You + a helper · about £195–300



1 · Buy

Qty	What	Where	£
2	**IKEA EKET wall cabinet, 35 cube, glass door** Two IKEA EKET wall cabinets with glass doors, 35 × 35 × 35, walnut effect, with their rails, and plugs and screws for a solid wall	IKEA	70–100
2	**Oak floating shelf, 900 × 230** Two oak floating shelves 900 × 230 with concealed brackets (B&Q Shelf Depot), for the window corner	B&Q	70–110
1 set	**Oak shelf over the door, two KUGGIS boxes** An oak shelf over the hall door, 91 × 27, on two brackets, and two IKEA KUGGIS lidded boxes 35 × 26 × 15, white	IKEA	35–60
2	**IKEA KUGGIS box, 35 × 26 × 15** Two IKEA KUGGIS lidded boxes, 35 × 26 × 15, white, for your black shelf	IKEA	20–30
	Total		**195–300**

2 · Cut

Solid oak, hardwax oil: each piece to scale



Part	What	Size (cm)	Qty	From	Notes
A	Door shelf	91 × 27 × 2.7	1	Solid oak	From the worktop's offcut; two brackets

Sizes are finished sizes, length × width × thickness; a 4 mm saw cut between pieces. Ply parts can be cut by the merchant from these layouts.

3 · Build

The wall cabinets

1. Mark the rail's line over the sofa's corner, the cabinets' bottoms at 162; detector first.
2. Screw IKEA's rail to the wall with 5 × 60 screws in brown plugs, level; hang the two EKET on it and tighten their fittings.

The alcove shelves

3. Mark the tops at 182 and 212 in the window corner; detector first.
4. Fix each concealed bracket with its own screws into plugs in the solid wall, check it's level, and slide the shelf on.

Over the hall door, and the boxes

5. The oak shelf's top at 208.7, on two brackets above the architrave (check it's no more than 7 wide).
6. Two KUGGIS boxes on it; two on your black shelf, either side of the laser, 1 over its front.

A · The network

The mesh up and the two switches, before any cables.

2 h · about £51–70

Buy

Qty	What	Where	£
11	**Cat6 patch lead, 0.5–2 m** Cat6 patch leads, 0.5–2 m	DIY shop or online	15–20
2	**TP-Link TL-SG108 8-port switch** TP-Link TL-SG108 8-port switch: one for the living room, one to replace the 5-port in bedroom 3	DIY shop or online	36–50
	Total		**51–70**

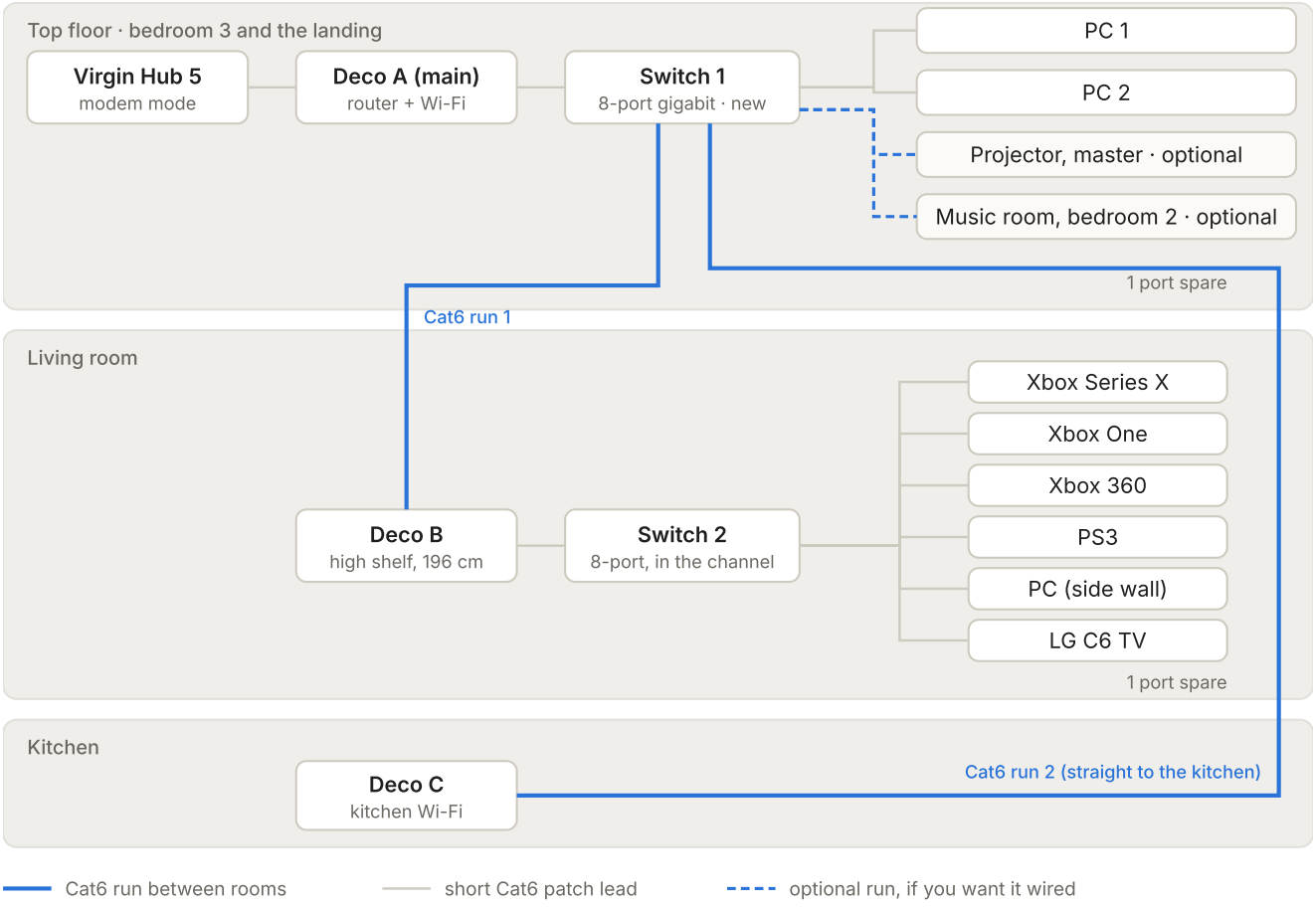
Do

Tools

- A phone with the Deco app
- A laptop for the hub's settings page

Every floor has its own Cat6 link back to bedroom 3

Deco A and Switch 1 live with the Hub 5 and the gaming PCs in bedroom 3; two spare ports can wire the projector and the music room



1. Unplug everything from the Hub 5. Log in at 192.168.0.1 with the password on its base, then Modem Mode → Enable Modem Mode → Apply. Its page moves to 192.168.100.1.

2. Connect a hub port to Deco A in bedroom 3 and set it up in the Deco app, reusing the old Wi-Fi name and password. If it shows no internet, try another hub port and restart the hub.
3. Plug the new 8-port Switch 1 into Deco A's other port, then the PCs into Switch 1.
4. Add Deco B in the living room and Deco C in the kitchen. For now they link over Wi-Fi.
5. Speed-test each room, and check the Xbox's network settings show NAT: Open.

Check before moving on: Internet in every room, and NAT: Open on the Xbox.

B · Cables and the socket

The Cat6 runs, the double socket and the PC's conduits: your own job, with the stop list.

2–3 days · about £165–345

Buy

Qty	What	Where	£
2–4 runs	**Cat6 solid copper cable** Cat6 solid pure copper, round (not CCA): two runs, four with the optional ones	DIY shop or online	30–60
1 set	**Cat6 tools: punch-down, keystones, tester, draw rods** Cat6 tools: a punch-down tool, RJ45 keystones and two surface boxes (or plugs and a crimper), a cable tester, a draw-rod set	DIY shop or online	30–60
1	**Double socket and back box** A double socket in the same finish, and a deeper back box if the old one is shallow	DIY shop or online	5–15
1 set	**Safety kit: GS38 tester, lock-off, socket tester** Electrical safety kit: a two-pole voltage tester to GS38 with its proving unit, an MCB lock-off and padlock, a plug-in socket tester	DIY shop or online	45–90
1 set	**Insulated screwdrivers, strippers, cutters** Insulated (VDE) screwdrivers, wire strippers and side cutters	DIY shop or online	20–40
1 set	**Oval conduit, 38 and 25 mm, and a 5 m C13 lead** Conduit: 38 mm and 25 mm oval, about 4 m each, two draw wires, one oak-coloured brush grommet; a 5 m C13 lead with a rewireable 13 A plug for the PC	Timber merchant	15–35
1	**Wall chaser for a day, and filler** For the chase: a wall chaser with dust extraction for a day (or a bolster chisel and club hammer), one-coat filler	DIY shop or online	20–45
	Total		**165–345**

Do

Tools

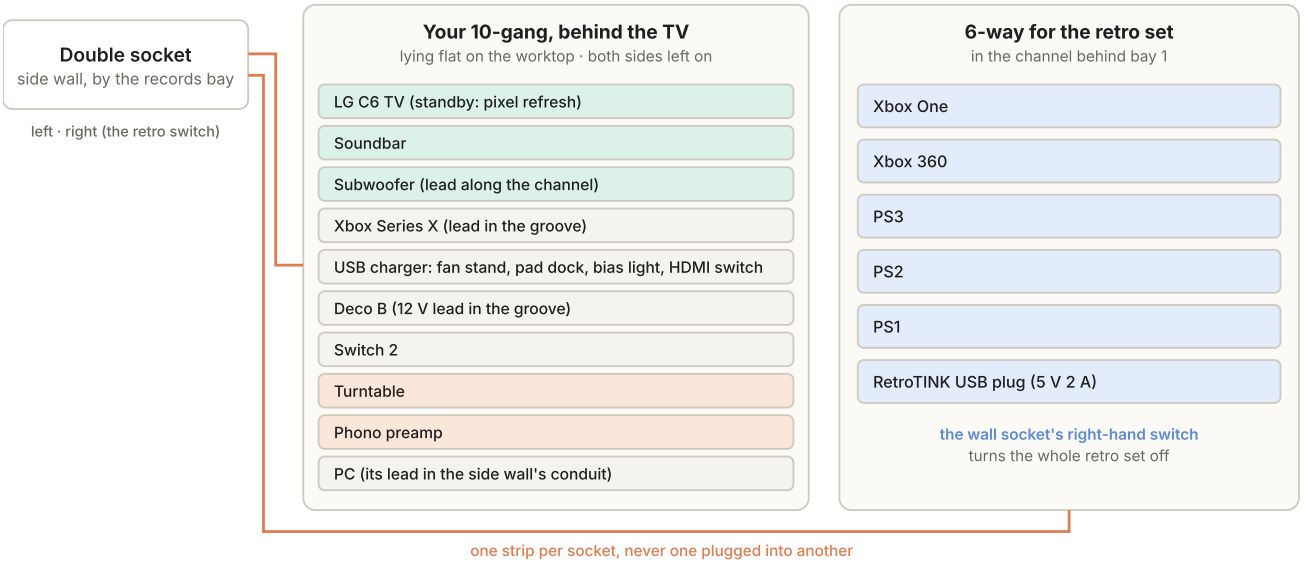
- Two-pole voltage tester (GS38) and its proving unit
- MCB lock-off and padlock
- Plug-in socket tester
- Insulated screwdrivers, wire strippers, side cutters
- Cat6 punch-down tool (or crimper) and a cable tester
- Draw rods
- Wall chaser with dust extraction, or a bolster and club hammer
- Cable and pipe detector
- SDS drill
- Safety glasses, ear defenders, FFP2 masks
- A helper

Fixings

- Cable clips for the Cat6 in the loft space or under the boards
- Brown wall plugs and 4 × 30 screws for the conduit clips

Power map: your 10-gang (now with the PC) plus a 6-way for the retro set

Everything modern on the left outlet, always on; the five retro consoles and the RetroTINK on the right, switched at the wall



- The socket on the side wall is a single today: you swap it for a double at the same spot (like for like, your own job: chapter B).
- The retro switch moves to the wall: with the TV 2 cm from the left upright, the 10-gang's own switches can't be reached any more.
- Rear speakers: your double socket behind the sofa's short side. Your lamp and the grow lights: the double behind the plant stand.
- The sub plugs into the 10-gang: its lead drops through the worktop's slot to the back of bay 2. Both spot bars use existing ceiling points.
- Load check: the 10-gang about 1.55 kW with everything on, PC included (850 W at most); the 6-way up to 0.8 kW. Each under 13 A (3 kW).
- Check the 10-gang is surge-protected; if not, use a surge-protected one for the TV and consoles. Label both outlets.

Before you start: the stop list

1. Stop and call an electrician if: the tester shows anything live after you've isolated; the wiring is old (rubber or fabric sheath, or red and black with no earth); there are more wires than you can match to your photo; there are scorch marks or the old socket is loose or hot; the consumer unit has rewireable fuses or no RCD; or the job grows into a new circuit.
2. Nothing in this step is notifiable in England: the socket is a like-for-like swap on an existing circuit, outside a bathroom. It must still be done to the wiring regulations, which is what this routine is for.

The Cat6 runs (low voltage: no isolation needed)

3. Measure each route on the house plan and add 2–3 m: bedroom 3, across the landing, down through the airing cupboard.
4. Lift a floorboard in the airing cupboard. Drill a 20 mm hole through the plate below (detector first), and feed a draw rod down to the alcove's right corner. Tape both Cat6 cables and a spare to it and pull them through. Pull the kitchen run on down the same way, past the right upright's groove.
5. Keep the Cat6 away from mains: cross at right angles, never run alongside in the same hole. Clip it every 30 cm; never staple through it or kink it.
6. Punch each end down onto a keystone in a surface box (or crimp a plug), to the T568B colours on both ends. Label both ends.
7. Test each run with the cable tester: all eight lights in order. Re-terminate any that don't.

Isolate the socket circuit

8. Tell everyone in the house you're working on the electrics.

9. At the consumer unit, switch off the breaker for the side-wall socket's circuit (usually a ring marked "sockets"). If the labels aren't clear, switch off the main switch instead.
10. Fit the MCB lock-off and padlock to that breaker, keep the key in your pocket, and tape a note on the board: "Working on this circuit, do not switch on".
11. Prove your voltage tester on its proving unit: it must light.
12. At the socket, with it pulled forward on its cables, test between every pair: live and neutral, live and earth, neutral and earth. No reading on any of them.
13. Prove the tester on the proving unit again. Only now is it safe to touch.

Swap the single socket for a double

14. Unscrew the old socket and ease it out. Photograph the wiring before you undo anything.
15. Check the stop list against what you see: an earth (green-and-yellow) wire present and sleeved, no scorching, the cable sheath into the box.
16. If the double needs a bigger box, chop the plaster out round the old box to the new box's outline and fit the new box with plugs and screws, the cables through its grommets.
17. Connect wire for wire: brown (or red) to L, blue (or black) to N, green-and-yellow to the earth terminal. Two cables in the old box go into the same terminals together. Strip only enough so no copper shows outside a terminal, and tighten every screw firmly.
18. Fold the cables in neatly without trapping them, and screw the socket on level.
19. Remove the lock-off, switch on, and plug the socket tester into each outlet: it should show correct wiring on both.

The PC's two conduits in the side wall

20. Mark the route from behind the PC shelf's back corner straight down to skirting height, then along behind the skirting to the base's plinth. Run the detector along it: move it if anything shows. Never chase near the socket's cables, which run straight up, down or across from it.
21. Chase two channels side by side, 5 cm apart: one for the 38 mm oval conduit (data: HDMI and Cat6), one for the 25 mm (the PC's mains lead). Vertical only on the wall, no deeper than the conduit needs. Wear the mask: the dust is silica.
22. Take the skirting off along the route, fix both conduits with a draw wire inside each, and refit the skirting over them.
23. Fill the chase with one-coat filler, sand flush, and photograph the route with a tape measure in the shot before you paint.

Finish

24. Mark every route on the plan, and keep the photos with it.

Check before moving on: The socket tester shows correct wiring on both outlets, both Cat6 runs test good end to end, and every route is photographed.

The wiring plan, before the carcass goes in

Lay the channel's trunking, the conduits and every lead below, label both ends, then build the base over them. Mains on one side of the trunking, HDMI, Cat6 and audio on the other.

Device	Leads, and where they plug in	Route
TV	mains → 10-gang (left outlet)	Behind the TV, on the worktop
Soundbar	mains → 10-gang (left outlet); HDMI eARC → TV HDMI 2	Along the worktop's back edge

Sub	mains → 10-gang (left outlet)	Down the slot behind the TV to the back of bay 2; keep its port clear
Series X	mains → 10-gang (left outlet); HDMI → TV HDMI 3	Down the groove behind the turntable-side upright
PC	mains (5 m C13) → 10-gang (left outlet); HDMI (active optical, 5 m) → TV HDMI 1; Cat6 → Switch 2	Brush grommet, then the two conduits chased into the side wall, behind the skirting to the plinth
Deco B · Switch 2	12 V → 10-gang (left outlet) · mains → 10-gang (left outlet); Cat6 from the landing	Deco B down the other upright's groove; Switch 2 on edge in the channel
Turntable · preamp	mains → 10-gang (left outlet); RCA → 3.5 mm (2 m) → soundbar AUX	The vinyl lead down the turntable-side groove
USB charger	mains → 10-gang (left outlet): fan stand, pad dock, bias light, HDMI switch	On the worktop behind the TV
Xbox One · Xbox 360 · PS3	mains → Retro 6-way (right outlet); HDMI → switch → TV HDMI 4	The channel's data side
PS2	mains → Retro 6-way (right outlet); component → RetroTINK; optical → soundbar	The channel
PS1 · RetroTINK	mains → Retro 6-way (right outlet) · USB 5 V 2 A → Retro 6-way (right outlet); SCART → RetroTINK → switch	The channel

C · Paint

Card Room Green on the walls and the skirting.

2–3 days · about £220–540

Buy

Qty	What	Where	£
1 lot	**Farrow & Ball Card Room Green: emulsion, eggshell, primer** Card Room Green: Estate (or Modern) Emulsion for walls, ceiling, alcove and chimney breast; Estate Eggshell for skirting and radiator (the door and its frame stay white); primer and undercoat	DIY shop or online	200–500
1 set	**Woodwork primer, sanding pads, sugar soap** Woodwork primer, sanding pads and sugar soap for the white skirting and door	DIY shop or online	20–40
	Total		**220–540**

Do

Tools

- Roller, tray and extension pole
- 25 and 50 mm brushes
- Dust sheets, masking tape
- Step ladder

1. Paint an A4 board in Card Room Green and look at it by day and with the LED strip on.
2. Wash the skirting and door with sugar soap, sand to a key, and prime them.
3. Two coats of emulsion on the walls, ceiling, alcove and chimney breast.
4. Two coats of eggshell in the same green on the skirting and radiator (the door and its frame stay white).
5. Leave it a week to cure before fixing anything to it.

Check before moving on: Two even coats, and no white showing at the edges or behind the radiator.

D · Lighting

The two spot bars and the LED strips.

1 day · about £288–795

Buy

Qty	What	Where	£
8	**Smart colour E14 bulb** Eight smart colour E14 bulbs for the bars: WiZ Colour P45 E14, 4.9 W (Wi-Fi, no hub), or Philips Hue White and Colour E14 lustre, 5.1 W, if a WiZ shows below its shade; both under the bar's 6 W	DIY shop or online	100–360
1 set	**Smart colour LED strip, 4.5 m, with channels** Smart colour strips for the upright grooves and under the side-wall run, in the same app as the bulbs: about 4.5 m of RGB and tunable-white strip (Philips Hue Lightstrip Plus, or WiZ's strip kits, cut at their marks), three aluminium channels with opal diffusers, plug-in power on the 10-gang (no fused spur); a 6500K bias light for the TV stays plain white	DIY shop or online	90–220
2	**Dunelm Wolston 4-light spot bar** Two Dunelm Elements Wolston 4-light spotlight bars, £49 each (bulbs above)	Dunelm	98–110
1	**Smart scene button (optional)** Optional: a smart button for the bars, so a scene is one press rather than the app (WiZ remote or Hue dimmer switch)	DIY shop or online	0–25
1	**Ceiling strip channel and diffuser (later)** Later: aluminium channel and opal diffuser for the existing ceiling LED strip (optional)	DIY shop or online	0–80
	Total		**288–795**

Do

Tools

- Two-pole voltage tester (GS38) and its proving unit
- MCB lock-off and padlock
- Insulated screwdrivers, wire strippers
- Step ladder
- A helper to hold the bar

Fixings

- Connector block or lever connectors (if the bar's flex needs them)
- Earth sleeving, insulating tape

Before you start

1. No dimmers: smart bulbs need full power, so leave the light switches as they are and keep them on. A dimmer in front of a smart bulb makes it flicker and can damage it.
2. The same stop list as chapter B. Here, also stop if a ceiling point has no earth (the bars need one), or the rose has more wires than you can match to your photo.

Isolate the lighting circuit

3. Tell everyone in the house you're working on the electrics.
4. At the consumer unit, switch off the breaker for the lighting circuit (usually marked "lights", 6 A). If the labels aren't clear, switch off the main switch instead.

5. Fit the MCB lock-off and padlock to that breaker, keep the key in your pocket, and tape a note on the board: "Working on this circuit, do not switch on".
6. Prove your voltage tester on its proving unit: it must light.
7. At each switch and each ceiling point, with the cover off, test between every pair: live and neutral, live and earth, neutral and earth. No reading on any of them.
8. Prove the tester on the proving unit again. Only now is it safe to touch.

Take down the old fittings, hang the bars

9. Unscrew each old fitting's cover. Prove dead again at its terminals. Photograph the wiring, mark the switched live (the wire from the switch) with tape, then disconnect and take the fitting down.
10. Fix bar 1's mounting plate to the ceiling over the rose between the TV and the sofa, the bar running from the TV towards the window, with screws into the joist or the plate's fixings. Bar 2's goes on the pendant point by the stairs, running the same way, along the west wall.
11. Connect earth first (green-and-yellow to the bar's earth), then neutral to N and the switched live to L, in the bar's terminal block; any loop wires go back together exactly as they were. Tighten, tuck, and fit the canopy.
12. Fit the smart colour bulbs and point every head straight down, evenly spread along the bar. Tilt one only if its pool lands in the TV's screen or someone's eyes. If a WiZ bulb shows below its shade's rim, use the shorter Hue lustre there.

Power on and test

13. Remove the lock-off and switch on, wall switches on. In the bulbs' app, add all eight, group them as bar 1 and bar 2, set warm white at about half, then save the three presets, Bright, Cosy and Film, from the lighting presets below (levels, and colours if you want them). Your red lamp's colour bulbs go in the same presets. With the table open, bar 2 comes up to at least 80 and bar 1 down to 15 (the table rule): save that as a fourth scene, Table, if your app can't do it on its own.

The low-voltage lights (nothing to isolate)

14. Cut the smart strip at its marks into three lengths: one for each upright's groove and one for the side-wall run. Press each into its aluminium channel under the opal diffuser; take the uprights' leads down their grooves to the strip's power on the 10-gang (smart colour).
15. The side-wall run's strip: in its channel in the groove under the run's front edge, 1.5 cm back from the edge, the full 220 cm; its lead drops with the PC's through the brush grommet into the conduit, to the same power.
16. Add the strips to the app beside the bars' bulbs, so each preset sets the wall's glow too: warm white for Bright, low and coloured for Cosy and Film.
17. Clip the ceiling strip through its notches in the uprights; stick the bias light round the back of the TV on its USB port.
18. Stand your red lamp in its spot (the 3D room shows the three), its bulbs green, pink and purple, top to bottom.

Check before moving on: Every bulb answers the app, the presets recall, and from the sofa no spot head shows in the screen or shines in your eyes.

E · Connect and finish

The TV, the consoles, the sound and the turntable, wired and tidied.

1 day · about £433–725

Buy

Qty	What	Where	£
1	**6-way surge-protected extension** 6-way surge-protected extension for the retro set (Xbox One, 360, PS3, PS2, PS1, RetroTINK)	Online	20–35
1	**4-port USB charger** 4-port USB charger, 5 V 2 A per port (HDMI switch, fan stand, pad dock, bias light)	DIY shop or online	10–20
1	**USB plug, 5 V 2 A** USB plug, 5 V 2 A or more, for the RetroTINK on the retro 6-way	Online	5–10
1 set	**Cable kit: grommets, velcro ties, labels** Cable kit: brush grommets, velcro ties, labels	DIY shop or online	20–30
1	**Flat trunking, 2 m** Flat two-compartment trunking for the 4 cm channel behind the base, about 2 m	DIY shop or online	10–20
1	**5-in-1 4K HDMI switch** 5-in-1 4K60 HDMI switch with remote	DIY shop or online	20–35
1	**Active optical HDMI cable, 5 m** Active optical HDMI cable for the PC, certified Ultra High Speed (48 Gbps), about 5 m	DIY shop or online	20–60
5 + 2	**HDMI lead, 1–2 m, and 90° adapters** High Speed HDMI leads, 1–2 m; two 90° HDMI adapters in case the TV's plugs need them	DIY shop or online	25–40
1	**RetroTINK-6X upscaler** RetroTINK-6X CE upscaler for the PS2 and PS1 (order on restock; UK price includes VAT and import fees)	Online	200–240
1	**PS2 component cable** PS2 component video cable, good quality	DIY shop or online	10–20
1	**Optical cable, 2 m** Optical (TOSLINK) cable, 2 m, PS2 straight to the soundbar	DIY shop or online	5–10
1	**PS1 RGB SCART cable** RGB SCART cable for the PS1	DIY shop or online	8–15
1	**Replacement stylus** Replacement stylus for the turntable (exact match for its model)	DIY shop or online	15–40
1	**Phono preamp (only if needed)** External phono preamp (only if the deck has a PHONO/LINE switch)	DIY shop or online	40–100
1	**RCA to 3.5 mm lead, 2 m** RCA-to-3.5 mm lead, 2 m: preamp under the Series X shelf to the soundbar's AUX	DIY shop or online	5–10
1 set	**Isolation feet and a record brush** Isolation feet for the turntable; a carbon-fibre record brush	DIY shop or online	20–40
	Total		**433–725**

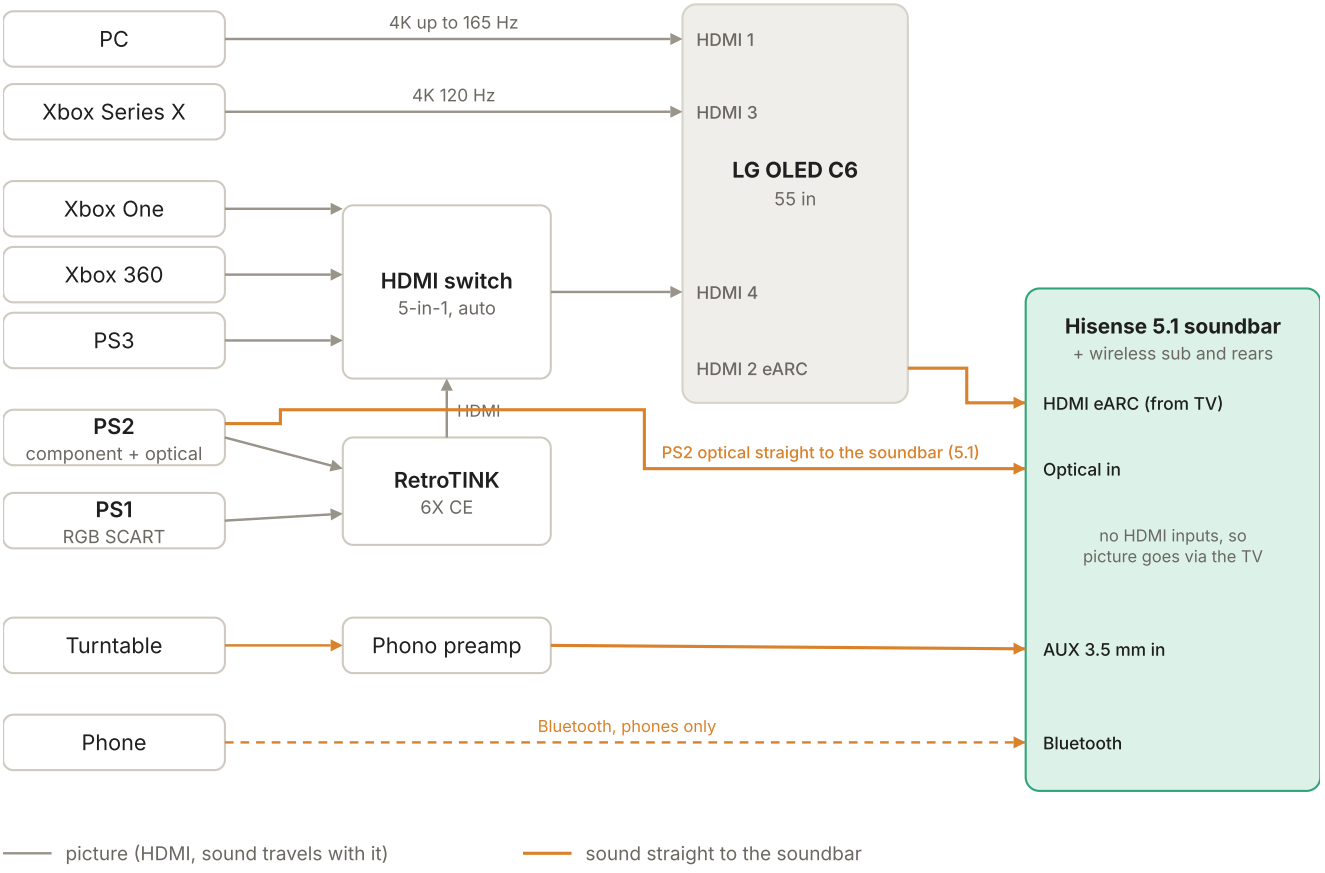
Do

Tools

- Phone with the Deco app
- Velcro ties

Everything sounds through the Hisense: directly where it can, via the TV where it can't

The soundbar has no HDMI inputs, so HDMI sources reach it over eARC; vinyl, PS2 audio and phones go straight in



1. TV inputs: HDMI 1 the PC, HDMI 2 the soundbar (eARC), HDMI 3 the Series X, HDMI 4 the HDMI switch (Xbox One, 360, PS3, RetroTINK).
2. The PS2 by component into the RetroTINK and by optical to the soundbar; the PS1 by RGB SCART.
3. The deck to the preamp, the preamp to the soundbar's AUX; set the tracking weight; soundbar in Music mode for vinyl.
4. Network stage 2: plug each Cat6 run into its Deco, check Deco B and C show "Ethernet", and give the Series X and PC high priority.
5. Label both outlets of the double socket and each HDMI switch input; set the sub and rear levels from the sofa.

Check before moving on: Everything on the network's "done when" list, and every source plays through the soundbar.

Good to know

Six techniques you'll use

1. **Fixing into brick.** Detect, mark, drill with the SDS on hammer setting, clear the dust, push in a brown (7 mm) wall plug and drive a 5 × 60 mm screw. In plasterboard, find a stud, or use heavy-duty plasterboard fixings rated for the load.
2. **Cutting oak cleanly.** Put masking tape over the cut line, score it with a knife, cut with the good face down (circular or track saw) or up (jigsaw). Clamp a straight edge as a guide.
3. **Routing a housing** (a shallow slot a shelf sits in). Clamp a straight batten as a guide, set the bit 4 mm deep, rout, then step down to 12 mm in passes. Practise on an offcut first.
4. **A wedged through-tenon.** Mark the 6 × 2.4 cm mortise on both faces of the upright and drill most of it out with a 20 mm bit from each side. Chisel it to the line. Saw the tenon on the shelf's end. Saw a kerf down its middle, glue it, tap it home, then drive a glued oak wedge into the kerf. When it's dry, saw it flush and plane it.
5. **Dowelling.** Use a dowel jig to drill matching 10 mm holes 3 cm deep in both parts, glue the dowels, clamp.
6. **Oiling.** Sand to 180 grit, wipe off the dust, rub on a thin coat of hardwax oil, wipe off the excess after 15 minutes, let it dry overnight, give it a second coat. Do the undersides too, so the boards don't cup.

Where beginners usually go wrong

- **Cutting the worktop to the tape instead of to a template.** Make a cardboard template of the alcove first; walls are rarely square.
- **Skipping the dry-fit.** Gluing a tenon that doesn't fit is the one mistake you can't undo.
- **Oil on one side only.** The board cups within a week.
- **Loading the LP crates before the anti-tip brackets go in.** A full crate weighs about 30 kg.
- **Folding the table up with something on it.** Clear it first, every time, and keep both turn buttons turned when it's up.

Tools

You don't need a workshop. Hire the two big tools for a weekend (a track saw and an SDS drill), or ask the timber merchant to cut the oak and ply to the cut lists: most will, for a small charge, and that removes the hardest cuts.

Tool	What it's for	Get it	About £
MEASURE AND MARK			
Tape measure (5 m) and a steel rule	every cut and every height	Buy	10–15
Combination square	squaring lines across boards and checking corners	Buy	10–15
Spirit level, 60 cm and 120 cm	levelling the base, the shelves and the table	Buy	15–30
Cross-line laser level	marking all the heights round the alcove in one go	Buy or borrow	30–60
Cable and pipe detector	checking every wall before you drill: non-negotiable	Buy	15–30
Pencil, masking tape, a marking knife	marking cuts (tape stops oak splintering)	Buy	5
CUT			
Track saw, or a circular saw with a straight-edge guide	the worktop, the table top and the long shelves	Hire or borrow	25–40 a day
Jigsaw	notches and the cable slot	Buy or borrow	40–70
Tenon saw (Japanese pull saw is easiest)	the tenons and the wedge kerfs	Buy	15–25

Sharp chisels, 6, 12 and 25 mm, and a mallet	cleaning out mortises and housings	Buy	30–50
DRILL AND FIX			
Cordless drill-driver (18 V)	pilot holes and screws; most of the job	Buy or borrow	60–120
SDS or hammer drill with masonry bits	plugs into the brick walls and the chimney breast	Hire or borrow	15–25 a day
Dowel jig and 10 mm brad-point bit	the dowels under the uprights and at the wrap ledge	Buy	20–35
Clamps: four 60 cm sash or F-clamps, four small	glue-ups and holding work while you fix it	Buy	40–60
Stud finder (if any wall is plasterboard)	finding studs to fix into	Buy	15
SHAPE AND FINISH			
Router (1/4 in is enough) with a straight bit and a guide	the housings, the LED grooves and the cable groove	Hire, borrow or buy	60–120
Random-orbit sander, 120 and 180 grit	smoothing the oak before oiling	Buy or borrow	40–70
Hardwax oil, lint-free cloths, a brush	finishing the oak	Buy	25–35
ELECTRICS (YOUR OWN: STEPS 3 AND 12)			
Two-pole voltage tester to GS38, with its proving unit	proving every wire dead before you touch it: non-negotiable	Buy	35–70
MCB lock-off and a small padlock	keeping the circuit off while you work on it	Buy	8–15
Plug-in socket tester	checking the new double socket is wired the right way round	Buy	8–15
Insulated (VDE) screwdrivers, wire strippers, side cutters	the socket and the ceiling fittings	Buy	20–40
Wall chaser with dust extraction (or a bolster and club hammer)	the PC's two conduits in the side wall	Hire	25–45 a day
Cat6 punch-down tool and a cable tester	terminating and checking the Cat6 runs	Buy	20–35
SAFETY			
Safety glasses, ear defenders, FFP2 dust masks	every cut, every sanding job; oak dust is an irritant	Buy	15–25
Work gloves; a first-aid kit	handling boards and sheet goods	Buy	10
A sturdy step ladder	the high shelf, the ceiling and the PC shelf	Borrow	—

Also on the list

Qty	What	Where	£
1	**Drip trays for the plant stand** Drip trays or a waterproof mat under the pots on the plant stand's slatted tiers; if you add clip-on grow lights later, a smart plug with a schedule	DIY shop or online	10–20
1 set	**Felt pads** Felt pads under the vinyl stand's feet, so it slides out from the breast without marking the floor	DIY shop or online	3–6

1	**Storage ottoman, navy velvet, 76 × 76 (ISTOOLS)** A storage ottoman in navy velvet, about 76 × 76 × 45 with a lift-off padded lid, in place of your blue cord one: ISTOOLS Large Velvet Storage Ottoman, 76 × 76 (check its height and price on the listing), or Dunelm's Square Opulent Velvet Storage Footstool in navy, 62 square, if you'd like it smaller	Dunelm	140–310
1	**Rug, about 230 × 160** A rug about 230 × 160 in front of the sofa: cream with deep blue, rust and ochre goes with the green, the oak and the blue sofa (the plan draws one as a guide)	Timber merchant	120–300
1 set	**Step drawers: ply boxes, oak fronts, runners** Drawers in the bottom three steps: three drawer boxes in 12 mm ply, 51 × 14.5 fronts in oak, three pairs of 350 mm runners, bearers under the treads	Timber merchant	180–320
1	**Blanket basket, water hyacinth, 59 × 51 × 29** The blanket basket at the sofa's open end: Dibor's Lakelyn water hyacinth lined basket, large, 59 × 51 × 29	DIY shop or online	36–42
	Total		**489–998**

For the timber merchant

Every part, by what it's cut from. The code is the chapter and the part's letter in it. Finished sizes in cm, length × width × thickness.

Solid oak, hardwax oil

Code	Part	Size (cm)	Qty
1A	Face frame	186 × 73 overall, 2 thick, stiles about 5 wide	1 set
1B	Drawer fronts	game drawer 56.8 × 6.5 × 2 · plinth drawer 54 × 6 × 2	6 · 12
1C	LP crate lips · slim drawer lips	36.3 × 5 × 1.5 · 16.5 × 5 × 1.5	2 · 3
1D	Follower blocks · Xbox One lift block	19 × 12 × 2 · 15 × 12 × 2	2 · 1
2A	Worktop	186 × 46 × 3	1
3A	Uprights	159 × 30 × 4.5	2
3B	High shelf, long length	128.9 × 30 × 4	1
3C	High shelf, short length	51.7 × 30 × 4	1
3D	Series X shelf	55 × 30 × 4	1
3E	Record ledge	128.9 × 7 × 2.4	1
3F	Ledge lips	128.9 × 1.5 × 0.9 · 126 × 1.5 × 0.9	1 · 1
3G	“Now playing” ledge	105.5 × 7 × 2.4, lip 1.5	1
3H	Wedges · pegs · dowels	6 × 3, tapering · 10 mm dowel, 5 long · 10 mm dowel	2 + spares · 6
4A	Side-wall run	220 long × 7 sweeping to 28 × 4, from one 225 × 30 × 4 board	1
4B	Keyboard tray housing	70 × 28 × 7, sides 7 × 2	1
5A	Table halves, centres	83.5 × 90 × 2.7	2
5B	Breadboard ends	90 × 7 × 2.7	4
5C	Apron rails	outer half 86.5 and 78 · inner half 84.9 and 78, all 12 × 42.6 × 2.2	2 + 2
5D	Table legs	6.5 × 4.5, cut 70 long	4
5E	Hinge rail	91 × 16.8 × 7	1
5F	Door stops (the cabinet)	8 × 2.5 × 1.2	2
6E	Bar counter	42.5 × 22.4 × 2.7	1
7B	Case sides	85 × 12 × 2	2
7C	Case top and bottom	86 × 12 × 2	2
8B	Record table lid	90 × 50 × 1.8	1
9A	Door shelf	91 × 27 × 2.7	1

Birch ply, 18 mm

Code	Part	Size (cm)	Qty
1E	Bay 3 sides	73 × 40	2
1F	Bay 3 top and bottom	56.4 × 40	2
1G	Plinth rails	120 × 9	2
1H	Plinth cross-pieces	34.4 × 9	3

Birch ply, 12 mm

Code	Part	Size (cm)	Qty
1I	Game drawer sides	35 × 15.5	2
1J	Game drawer back · front panel	51.9 × 15.5 · 51.9 × 6	1 · 1
1K	LP crate bases · slim drawer bases	34.9 × 40 · 15.3 × 40	2 · 3
1L	Bay 3 divider	69.4 × 38	1

1M	Plinth drawer sides · ends	34 × 4.4 · 49.6 × 4.4	2 · 2
1N	Bay 2 divider · part shelf	35.3 × 38 · 32.2 × 36	1 · 1
4C	Keyboard tray	54 × 26	1
8A	Record table base	84 × 41	1

Birch ply, 9 mm

Code	Part	Size (cm)	Qty
1O	Game drawer base	52.5 × 34.4	1
1P	LP crate sides · ends	40 × 8 · 33.1 × 8	4 · 4

Birch ply, 6 mm

Code	Part	Size (cm)	Qty
1Q	Game drawer cross divider	51.9 × 9	1
1R	Game drawer bin walls · bin floors	12.6 × 9 · 16 × 12.6	2 · 2
1S	Plinth drawer base	52 × 34	1
1T	Slim drawer sides · backs and fronts	40 × 10 · 14.1 × 10	6 · 6
5K	Cabinet backs	91 · 49, both × 184.2	1 · 1
6D	Bar back	38.9 × 82.5	1

Oak-faced ply, 18 mm (the dining cabinet and the bar)

Code	Part	Size (cm)	Qty
5G	Cabinet ends	189 × 23.4	4
5H	Cabinet tops · floors	91 × 23.4 · 49 × 23.4	2 · 2
5I	Cabinet linen shelf · chair shelf	91 × 16.8 · 49 × 20.8	1 · 1
5J	Cabinet doors	90.5 · 48.5, both × 183.8	1 · 1
6A	Bar ends	87.3 × 22	2
6B	Bar floor · shelf	38.9 × 22 · 38.9 × 19.6	1 · 1
6C	Bar doors · plinth front	82.2 × 19.25 · 38.9 × 3	2 · 1

Oak-faced ply, 6 mm

Code	Part	Size (cm)	Qty
7A	Glass case back	86 × 81	1

Sheets to buy, nested

Every ply part laid out on the fewest boards (a full sheet is 244 × 122; half and quarter sheets where a full one would be mostly offcut), a 4 mm saw cut between parts, grain along the long side. Hand these to the merchant with the cutting list.

Sheet	Boards (cm)	Parts	Used	Biggest offcuts, and what for
Oak-faced ply, 18 mm (the dining cabinet and the bar)	3 × 244 × 122	19	68%	244 × 38.95: a spare shelf or a drawer box; 152.6 × 16.8: drawer sides, cleats or jigs
Oak-faced ply, 6 mm	1 × 122 × 122	1	47%	122 × 40.6: a spare shelf or a drawer box; 35.6 × 81: a spare shelf or a drawer box
Birch ply, 18 mm	3 × 122 × 61	9	60%	122 × 20.6: drawer sides, cleats or jigs; 122 × 11.2: packers, shims and test cuts
Birch ply, 12 mm	1 × 244 × 122	18	59%	244 × 24.6: drawer sides, cleats or jigs; 25.1 × 34.9: packers, shims and test cuts

Birch ply, 9 mm	1 × 122 × 61	9	56%	28.7 × 34.4: packers, shims and test cuts; 88.5 × 8: packers, shims and test cuts
Birch ply, 6 mm	2 × 244 × 122	21	59%	244 × 33.3: a spare shelf or a drawer box; 161.1 × 38.9: a spare shelf or a drawer box